Proximity to the Plague: Measuring Locust Exposure and Child Health*

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Abstract

I investigate the early childhood health outcomes of the Kenyan 2020–2021 desert locust invasion. To account for the multi-harvest setting, I build a treatment that conditions on the insect's biological capacity to damage vegetation, the growth stage of the local crop, and subsequent vegetal decline, matching georeferenced locust observations to the 2022 Kenya Demographic and Health Survey. Children exposed in utero within 30 km of a verified damage event carry weight-for-height z-scores 0.185 standard deviations below the control group, and are 26 percent more likely to be clinically wasted. Effects are strongest when children are exposed in the second and third trimester. Riverine livelihood zones, where households depend heavily on their own harvest and have the least market cushion lose an additional 0.25 standard deviations on weight-for-height beyond the average cropping-zone effect. The results suggest that the health consequences of an agricultural shock hinge on substitution as the key margin of adjustment, with households least able to replace lost harvests through markets bearing the largest nutritional penalty.

Keywords: desert locust, child health, wasting, exposure measurement, Kenya

JEL Codes: I15, O12, Q12, Q54, C18

*All errors are my own.

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1 Introduction

Natural disasters that affect agricultural production often have adverse nutritional effects for young children in low- and middle-income countries. The harsh conditions that these children face in early childhood compound to detrimental effects later in adulthood (Almond and Currie, 2011; Maccini and Yang, 2009; Stein and Susser, 1975). To better coordinate policy efforts, researchers have attempted to identify the effects of these agricultural disruptions on early childhood health using a variety of methods.

Cross-sectional household surveys such as the Demographic and Health Surveys (DHS) and the Multiple Indicator Cluster Surveys (MICS) are widely used to identify the consequences of natural disasters (Cooper et al., 2019; C. and Rieger, 2024; Persson et al., 2012; Sajid and Bevis, 2021; Daoud et al., 2016). As these surveys provide geocoded sample clusters, a researcher can link a household to the shock of interest by drawing a radius around the observed event and treating everyone inside this radial "buffer" as exposed.

A small but growing literature in economics has come to employ these techniques to the incidence of locust plagues (Chatterjee, 2022; Le and Nguyen, 2022; De Vreyer et al., 2015). These papers often assign exposure from a locust's recorded presence within a radius or an administrative boundary, without conditioning on the insect's biological capacity to harm vegetation, the phase of a crop's life cycle or on whether a crop was subsequently damaged following exposure to the plagues. Conte et al. (2023) use a proximity-based measure and rely on Mali's primarily single harvest to time specific pathways to health from the locusts. However, many countries hit by these types of agricultural shocks are bimodal; they have multiple harvests that follow from two rainy seasons in a year.

This distinction might affect the outcome and corresponding policy implications. A second season might let a household replant and recover. However, the stocks from one harvest are only meant to last until the next, so a household that loses a crop may have little to fall back on. If replanting lets households recover, relief can be brief and limited to seed and inputs for the next season. If stocks run out first, households need food or cash to carry them to the next harvest.

The 2020-2021 desert locust invasion mainly hit bimodal East African countries such as Uganda, Ethiopia, Somalia and Kenya. I estimate the effect on early childhood health from this crisis in Kenya.

I match georeferenced locust observations to the 2022 Kenya DHS and build an exposure measure that requires the insect to have arrived when a crop was actually standing in the field. Kenya's bimodal rainfall gives it two planting windows, two harvests, and crop calendars that differ by several months across the 17 of its 24 livelihood zones that grow crops. This means that locusts that land on two different areas on the same date might feed on susceptible crops in one zone and fly over bare ground in the other. I use that calendar to gate exposure: a locust record counts only if it falls within a given radius of a survey cluster while the child was in utero, if it describes a gregarious form capable of stripping a field, if the local crop calendar places a crop in a susceptible growth stage on the record's date, and if satellite vegetation over the cluster then declines relative to that cluster's own pre-invasion climatology. Due to the geographical displacement of the DHS, I build the measure at nine radii from 10 to 50 km.

Children exposed in utero under this measure carry weight-for-height z-scores 0.185 standard deviations below unexposed children in the same survey cluster born in a different month. Adding county-specific linear time trends raises the estimate to 0.207. At the clinical threshold the probability of wasting rises 1.9 percentage points against a baseline of 7.2 percent, a relative increase of 26 percent, and 3.6 percentage points once county trends are absorbed. The estimate is stable across the child, family, and trend specifications and is significant at the 1 percent level in every one. It is negative and significant at the 10 percent level or better at every radius from 10 to 50 km and generally grows as the radius tightens, reaching 0.313 standard

deviations within 10 km.

Wasting records a deficit of weight relative to attained height and responds to adverse nutritional effects in the short term within months. Height-for-age records accumulated growth over the whole of a child's life and moves slowly. The median child in this sample was 14 months old at measurement, so a disruption confined to gestation and the months around birth has had little time to register in linear growth. Height-for-age z-scores are significant at 30 km when exposed only in the third trimester. The effects on wasting are strongest and significant when children are exposed in the second and third trimester. They are also concentrated among children born after the first verified swarm reached their area but before the affected harvest was due, and local maize prices do not account for them.

These results suggest that relief should go to households whose standing crop was destroyed rather than to everyone near a swarm, should reach women in late pregnancy, and should arrive between the swarm and the failed harvest, in the form of food or cash rather than measures aimed at local prices.

The paper also contributes to the fetal origins literature. Barker (1990) proposed that adult disease traces to conditions in-utero, and economists have since documented effects of prenatal shocks on long-term health, schooling, and earnings (Almond and Currie, 2011; Almond et al., 2018). Rainfall and drought supply most of the identifying variation. Maccini and Yang (2009) link early-life rainfall to adult height and schooling in Indonesia, Hoddinott and Kinsey (2001) and Alderman et al. (2006) document persistent growth faltering after Zimbabwe's droughts, and Shah and Steinberg (2017) show that rainfall raises human capital in utero while lowering it at school age. Anthropometric z-scores are the standard outcome because early childhood growth predicts adult human capital (Victora et al., 2008). Rainfall shocks identify the effect of income and food availability on the fetus. But they cannot separate the agricultural channel from the waterborne-disease channel, the maternal-labor channel, or the price channel. My exposure measure isolates the agricultural channel by requiring a damaging swarm, a crop in the ground, and a subsequent decline in vegetation.

The remainder of this paper is organized as follows. Section 2 describes the locust, Kenya's agricultural landscape, and the 2020–2021 invasion. Section 3 describes the data. Section 4 builds the exposure measure and the estimating equation. Section 5 reports results, robustness, mechanisms (Section 5.4) and heterogeneity (Section 5.5). Section 6 discusses the results.

2 Setting

2.1 Kenya's Agricultural Landscape

Kenyan agriculture is organized around two rainy seasons. The long rains fall from roughly March to May and the short rains from October to December. Most of the country's maize, the dominant food staple, is planted with the long rains in the western and Rift Valley highlands and harvested from August onward. Eastern and coastal areas depend more heavily on the short rains and on drought-tolerant crops including sorghum, millet, and cowpeas. The timing of planting, vegetative growth, and harvest therefore differs across the country by several months, and a locust arrival that is catastrophic in one county may be harmless in another on the same date.

County boundaries do not track how households feed themselves. Figure 1 maps the Famine Early Warning System Network's (FEWS NET) 24 livelihood zones under the household economy approach, each covering an area whose residents share a production system and a set of food-access strategies (Famine Early Warning Systems Network (FEWS NET), 2011).

Cropping zones cover the western and central highlands, the eastern slope of the highlands, and the coastal strip. They include the Central Highlands High Potential Zone (KE19), the Western Medium and High Potential Zones (KE20 and KE21), the Southeastern mixed farming zones (KE16 and KE17), two coastal farming zones (KE12 and KE13) and two marginal mixed farming zones (KE06 and KE22). Households in these zones grow maize, beans, potatoes, and in the highlands tea and coffee, and buy the balance of their food.

Pastoral zones cover the arid north and northeast and a belt of southern rangeland. They include the Northwestern, Northern, Northeastern, Eastern, Southeastern, and Southern Pastoral Zones (KE01, KE05, KE09, KE10, KE11, KE15). Households herd camels, cattle, and small stock, move with the rains, and obtain food from milk, livestock sales, and purchase.

Agropastoral zones sit on the transition between the two and combine rain-fed cropping with herding (KE03, KE07, KE18, KE24). Riverine zones are narrow irrigated strips along the Turkwell, the Dawa at Mandera, and the Tana (KE02, KE08, KE14). Fishing zones cover the shores of Lake Turkana and Lake Victoria (KE04, KE23).

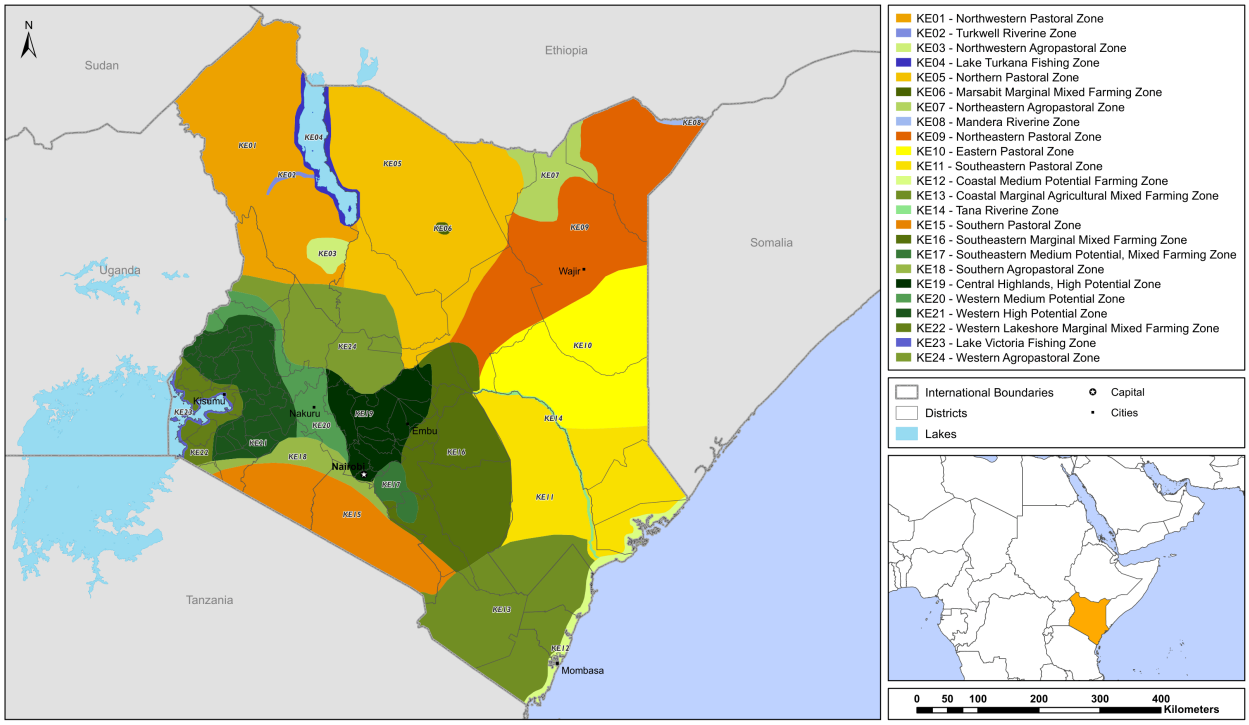


Figure 1: FEWS NET livelihood zones of Kenya. Source: (Famine Early Warning Systems Network (FEWS NET), 2011)

2.2 The Desert Locust

The desert locust, *Schistocerca gregaria*, is a grasshopper that changes phase with population density. At low densities it is solitary (solitarious). These locusts avoid one another, move relatively little, and typically remain within a restricted area, dispersing rather than congregating. Their coloration is usually cryptic, matching the background vegetation and soil, and their bodies are slimmer with proportionally longer legs. Because they neither aggregate nor migrate in large numbers, solitary locusts cause little or no damage to

crops and are not considered a plague threat. This phase persists as long as densities remain low, and it is the state in which locust populations spend most of their time.

In the gregarious phase, by contrast, locusts are crowded and highly social. When rainfall concentrates individuals onto scarce green vegetation, crowding triggers a transformation toward this phase over hours to generations (Symmons and Cressman, 2001). Gregarious individuals aggregate, march as hopper bands before they can fly, and form adult swarms that travel with the wind over hundreds of kilometers. They develop striking black-and-yellow (or pink in immature adults) warning coloration, broader bodies, and shorter legs, and they are more active and voracious than their solitary counterparts. Swarms can contain billions of individuals and consume vast quantities of vegetation in a single day, making the gregarious phase the one responsible for the devastating agricultural damage for which desert locusts are notorious. The two phases are endpoints of a continuous, density-dependent continuum, with a transient (transiens) phase occupying the middle.

Only gregarious swarms and hopper bands reach the densities that strip a field. Solitarious individuals occur throughout the locust's recession area in most years without agronomic consequence.

Damage also depends on timing. Locusts are polyphagous and consume cereals, legumes, and pasture, but the loss a field sustains depends on what stage the crop has reached. Standing biomass close to harvest cannot be replaced within a season. A field at planting can be resown at the cost of seed and labor. Grain already harvested and stored is not at risk, and a field out of season has nothing to lose. A swarm passing over a harvested field is agronomically irrelevant even though it appears in the record as a locust observation.

Upsurges are episodic. Global control efforts suppressed them for much of the late twentieth century (Zhang et al., 2019), and recent work links renewed activity to extreme rainfall events and their climatic drivers (Salih et al., 2020; Liu et al., 2024; Peng et al., 2020).

2.3 The 2020–2021 Kenyan Crisis

Heavy and repeated rainfall over the Arabian Peninsula in 2018 and 2019 produced several undetected breeding generations of the desert locust. Tropical cyclones delivered exceptional precipitation to usually arid terrain, creating temporary lakes in the desert and allowing locusts to breed (Salih et al., 2020). These favorable conditions sustained three generations of breeding in roughly nine months.

Swarms then crossed into the Horn of Africa through Yemen, Somalia, and Ethiopia, reaching northern Kenya in late December 2019. By early 2020 swarms had spread across much of central and eastern Kenya, prompting the national government to declare an unprecedented threat to food security and livelihoods. The invasion represented the worst locust upsurge Kenya had experienced in seventy years.

The upsurge continued into 2021. A second wave entered Kenya from Ethiopia and Somalia in mid-November 2020 and eventually reached 15 counties, reversing the gains of the preceding months (Ministry of Agriculture, Livestock, Fisheries and Cooperatives, Kenya, 2021). Kenya responded with nine surveillance and spray aircraft, 21 ground spray vehicles, and more than 500 trained personnel, operating alongside FAO-supported control operations (EnviroNews Nigeria, 2021). Ground and aerial spraying proceeded while breeding continued, and control teams worked to suppress new hopper bands before they could fledge into swarms. The campaign brought the invasion largely under control by mid 2021 with some gregarious sightings in the fall of that year. The FAO declared the invasion officially over in February 2022 (Food and Agriculture Organization of the United Nations (FAO), 2022b).

Figure 2 plots daily counts of gregarious swarm and hopper-band observations across the episode. The invasion arrives in two waves. The first begins in late December 2019, peaks in May and June 2020, and

subsides by August. The second builds from December 2020, peaks in February 2021, and ends in April. The shock therefore ran for sixteen months and crossed two long-rains seasons and one short-rains season.

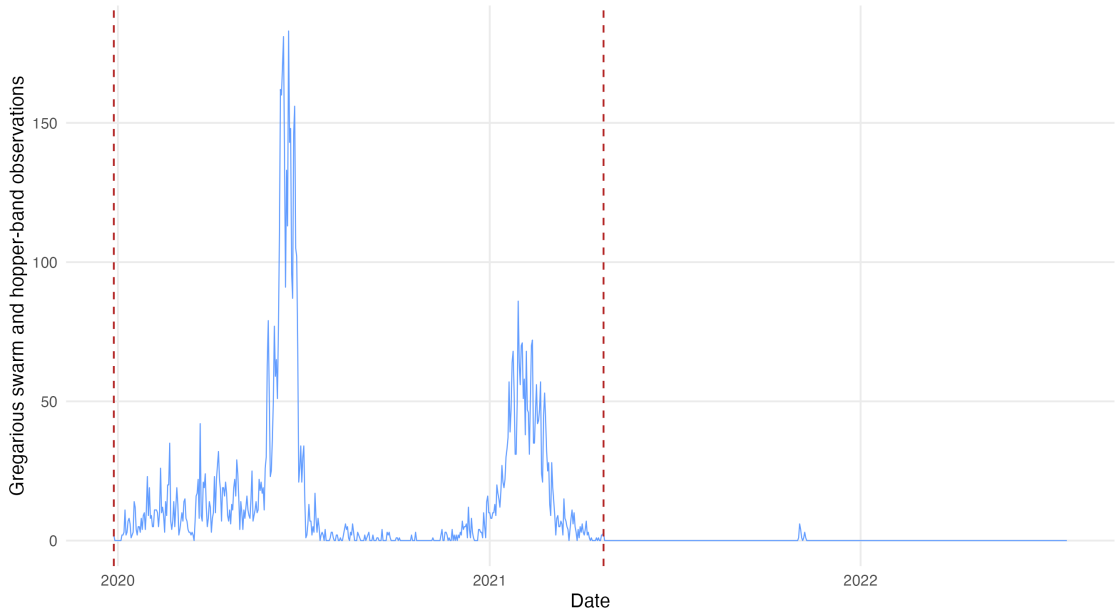


Figure 2: Daily gregarious swarm and hopper-band observations in Kenya. Dashed lines mark the derived invasion window.

3 Data

3.1 Household Survey

The child-level sample is drawn from the 2022 Kenya Demographic and Health Survey (DHS), with the 2014 round used as a pre-invasion falsification sample (Kenya National Bureau of Statistics (KNBS) and ICF, 2023; Kenya National Bureau of Statistics (KNBS) and ICF International, 2015). The DHS is a nationally representative household survey conducted under the DHS Program (ICF, formerly funded principally by USAID), fielded in Kenya roughly every five years. Each round enumerates all children born to sampled women in the preceding five years, with retrospective birth histories, anthropometric measurement (height and weight, converted to age- and sex-standardized z-scores against the WHO growth reference of WHO Multicentre Growth Reference Study Group (2006)), and household, maternal, and child covariates.

Each DHS cluster (an enumeration area, roughly a village or urban block) is assigned a single GPS coordinate, randomly displaced by up to 2 km for urban clusters and 5 km for rural clusters (up to 10 km for 1 percent of rural clusters, restricted to stay within the same second administrative region) to protect respondent confidentiality (Burgert et al., 2013). Raster and vector extraction in this paper such as rainfall and distance to roads, is performed as a buffer mean or nearest-feature match over this displacement radius rather than at the point coordinate.

The 7,603 children with valid anthropometry are spread across 1,644 clusters, a mean of 4.6 children per cluster and a median of 4, ranging from a single measured child in the thinnest clusters to 19 in the largest. The sample is restricted to children whose gestation overlapped the invasion window, 28 December 2019 to

23 April 2021. These children were born between December 2019 and January 2022 and were 1 to 30 months old at measurement.

3.2 Locust Records

Locust observation records come from the Reconnaissance and Management System of the Environment of Schistocerca (RAMSES), the geographic information system used by national Locust Information Officers in each affected country to record survey and control activity, and consolidated by the Food and Agriculture Organization's Desert Locust Information Service (DLIS) (Food and Agriculture Organization of the United Nations (FAO), 2022a). Records are distributed through the FAO Locust Hub. Each record carries a date, coordinates, a presence type (e.g. hopper band, adult swarm, or an explicit absence), a phase classification (gregarious, transiens, or solitarious), a maturity classification, and, where control operations occurred, the control type.[1]

3.3 Climate and Vegetation

I use three remotely sensed products: CHIRPS rainfall, TerraClimate potential evapotranspiration, which I combine with CHIRPS into the SPEI drought index, and MODIS NDVI. Rainfall comes from the Climate Hazards group InfraRed Precipitation with Station data (CHIRPS), a quasi-global, 0.05°-resolution, daily-to-monthly precipitation dataset combining infrared satellite imagery with station observations, produced by the Climate Hazards Center at UC Santa Barbara and designed for drought monitoring in data-sparse regions (Funk et al., 2015). Both monthly and dekadal (10-day) CHIRPS products are used: the monthly product supports gestational rainfall controls and the drought construction described next, and the dekadal product supports growing-season onset detection for the crop calendar.

The Standardized Precipitation Evapotranspiration Index (SPEI), at 6-month accumulation windows, is constructed from CHIRPS precipitation and potential evapotranspiration (PET) from TerraClimate, a global, ∼4 km monthly climate and water-balance dataset (Abatzoglou et al., 2018). SPEI standardizes the climatic water balance (precipitation minus PET) rather than precipitation alone, following the index definition in Vicente-Serrano et al. (2010); because the drought conditions overlapping the invasion period combined rainfall deficits with elevated evaporative demand, a precipitation-only index might understate its effects.

Vegetation condition comes from MODIS/Terra Vegetation Indices (MOD13Q1), a 16-day, 250 m composite Normalized Difference Vegetation Index (NDVI) product from NASA's MODIS sensor, accessed via the AppEEARS extraction service (Didan, 2015). Pixels are screened on the product's reliability flag before averaging, so cloud- or snow-contaminated observations never enter a cluster mean. NDVI levels are converted to per-cluster, per-composite z-scores against a pre-invasion climatology.

3.4 Crop Calendars

The exposure measure requires knowing, for a given cluster on a given date, whether a crop was standing in a field and how far it had advanced. Livelihood zone boundaries come from shape files that FEWS NET provides; the FEWS NET Kenya livelihood zone map described in Section 2 (Famine Early Warning Systems Network (FEWS NET), 2011).

[1] Unfortunately, control operation data are sparse and recorded with many errors in this sample.

The FEWS NET livelihood profiles also published a seasonal calendar for each of Kenya's 24 livelihood zones, showing land preparation, planting, and harvesting by crop alongside livestock, labor, and market-access cycles (Famine Early Warning Systems Network (FEWS NET), 2011). A cropping zone's calendar lists maize, beans, potatoes, and the relevant cash crops. A pastoral zone's calendar lists calving, kidding, migration, and milk availability, and contains no crop row. Clusters in zones without a crop row are never placed in a susceptible stage, so they can never satisfy the stage condition.

Each seasonal calendar uses a three-color Gantt legend, land preparation, planting, and harvest, plotted month by month for every crop grown in that zone (Famine Early Warning Systems Network (FEWS NET), 2011).[2] An automated pixel-extraction routine locates the table grid and the legend's three reference colors on each of the 24 zone pages, then classifies every crop row's bar by nearest-color match at semi-month resolution. Seventeen zones return at least one crop row. The other seven are auto-detected as having no crop calendar at all, these seven are six pastoral plus the Lake Turkana fishing zone.[3] The automated output was checked against an independent manual pixel read of one page, which caught and fixed an off-by-one error in which row's bar was attributed to which crop, and all 21 automatically read zones run through the identical code path after that fix.

The three-color legend collapses to a single susceptibility ranking. Land preparation leaves bare ground and is not susceptible. Planting is moderately susceptible, since a destroyed planting can be resown at the cost of seed and labor. Harvest, when a mature crop stands in the field, is the most susceptible stage. A cluster-date outside every crop's bar on its zone's page is out of season. The calendars leave the months between the end of planting and the start of harvest blank, so a crop that is growing in those months counts as out of season (Appendix B.3).

Where a cluster has more than one crop group or more than one season active on a date, the governing stage is the most susceptible of those active, as Section 4 formalizes this.

3.5 Food Prices

Prices come from the World Bank's Real-Time Prices (RTP) platform, which supplies a monthly price estimate for maize and beans at 233 Kenyan markets from 2007 onward, combining sparse directly-observed transactions with a model that imputes missing values from nearby markets, related commodities, and historical patterns, following the method in Andrée (2021). I take the platform's own "close" estimate, its best and most current price for a given market-month, as given rather than re-deriving it; I restrict the sample to 2015–2022.

3.6 Roads

Distance from each DHS cluster to the nearest major road is computed from OpenStreetMap's Kenya road network, exported through the Humanitarian OpenStreetMap Team's export tool (Humanitarian OpenStreetMap Team (HOT) and OpenStreetMap contributors, 2022). This is a crowd-sourced, continuously updated vector road network.

[2] Figures 14 and 15 in Appendix B provide examples of one calendar for each type.

[3] Three zones, Tana Riverine, Western High Potential, and Lake Victoria Fishing, needed the corresponding page read by hand instead: the automated color-distance threshold that separates the legend colors was not stable across the environments the pipeline was run in for these three pages specifically, so rather than tune a threshold, those three pages are transcribed directly and flagged as such in the underlying data.

3.7 Administrative Boundaries

County boundaries come from the geoBoundaries Global Database of Political Administrative Boundaries, distributed via the Humanitarian Data Exchange (HDX) as the Kenya subnational administrative boundaries dataset. Kenya's 47 counties correspond to the ADM1 level in this dataset, which is produced and maintained by geoBoundaries and published under an open license (geoBoundaries, 2022).

4 Empirical Strategy

4.1 Exposure construction

Let $\mathcal{L}$ denote the set of locust monitoring records, where each record $\ell \in \mathcal{L}$ carries a location, an observation date δ_ℓ, a presence indicator $P_\ell \in \{0, 1\}$, and a phase $\phi_\ell \in \{\text{gregarious}, \text{solitarious/transiens}\}$.[4] A record's *form* is

$$\text{Form}(\ell) = \begin{cases} \text{damaging} & \text{if } P_\ell = 1 \text{ and } \phi_\ell = \text{gregarious} \\ \text{benign} & \text{if } P_\ell = 1 \text{ and } \phi_\ell = \text{solitarious/transiens} \\ \text{absent} & \text{if } P_\ell = 0. \end{cases} \tag{1}$$

Damaging records are present records in the gregarious phase: swarms, hopper bands, and a small number of adult and hopper groups. Benign records are present records in the solitary or transiens phase, whatever their presence type, and are retained throughout as the falsification group in Section 5.3.

For child i born at date b_i to survey cluster $c(i)$, the gestational window is $W_i = [b_i - 266,\ b_i]$ under an assumed 266-day gestation.[5] For a radius r, the set of records a child could have been exposed to is

$$\mathcal{E}_i(r) = \{\ell \in \mathcal{L} : \text{dist}(c(i), \ell) \le r,\ \delta_\ell \in W_i\}, \tag{2}$$

the intersection of geographic proximity and gestational timing. Exposure is indexed by child rather than by cluster, since two children in the same cluster born months apart have different windows W_i and can differ in exposure holding cluster and radius fixed.

4.1.1 Form

The first condition is that the record describe a locust at a density capable of stripping a field. This uses the phase classification already coded in the monitoring data through equation (1). Records that are excluded are carried through the identical construction below as $\text{Form}(\ell) = \text{benign}$ and used as a falsification group.

[4] RAMSES leaves the phase field blank for 83 present records, about 1 percent of the records I classify as gregarious, and I assign them to the gregarious phase. All 83 are confirmed observations made between March and December 2020, during the first wave of the invasion. Of the 48 that carry an observer's comment beyond a source tag, 21 describe swarms, hopper bands, crop destruction or a heavy infestation, 12 describe spraying or a call for control, 10 describe flying, roosting or settled adults, one describes scattered adults feeding lightly, one names a tree locust, and three, one of them a test entry, give no information on phase. Classifying these records as non-damaging instead changes the treatment status of 5 of the 1,703 treated children in the weight-for-height estimation sample at the 30 km radius and moves the weight-for-height estimate by less than 0.01 standard deviations.

[5] Tables 6 to 8 in Appendix A vary each choice in the exposure construction one at a time, at the 10, 30 and 50 km radii. A 280-day gestation, counted from the last menstrual period, gives a weight-for-height estimate of −0.159 at 30 km, significant at the 1 percent level, against −0.185 under the main definition. At 50 km it falls to −0.091 and loses significance.

4.1.2 Crop stage

The invasion struck at different times across the agricultural calendar, and its timing varied from place to place. The May and June 2020 peak fell while the long-rains maize crop stood under its harvest bar in the western zones and was still at planting in the Central Highlands, while the short-rains crop stood under its own harvest bar in parts of the east, and over bare ground under land preparation in counties between seasons. The February 2021 peak fell after the short-rains harvest bar had closed in most of the country. As calendars differ across Kenya by several months, an arrival date on its own carries no information about whether a crop was lost. Two children conceived in the same month, in clusters 200 km apart, can face swarms on the same dates and face different agronomic consequences.

Thus, the second condition is that the arrival find a crop worth destroying. Let $\mathcal{S}(c, \delta)$ be the set of crop stages active in cluster c's crop calendar on date δ, possibly more than one under staggered planting or multiple crop groups, and possibly empty. Each stage carries a susceptibility rank,

$$\text{rank}(s) = \begin{cases} 3 & s = \text{harvest (standing crop)} \\ 2 & s = \text{planting} \\ 1 & s \in \{\text{land preparation}, \text{out-of-season}\}, \end{cases} \tag{3}$$

and the governing stage on a cluster-date is the highest-ranked active stage,

$$S(c, \delta) = \underset{s \in \mathcal{S}(c,\delta)}{\arg\max}\, \text{rank}(s), \qquad S(c, \delta) = \text{out-of-season if } \mathcal{S}(c, \delta) = \varnothing. \tag{4}$$

A stage is susceptible if its rank exceeds 1,

$$\text{Susceptible}(s) = \mathbb{1}[\text{rank}(s) \geq 2]. \tag{5}$$

The harvest stage, when a mature crop stands in the field, is susceptible at rank 3. Planting is susceptible at rank 2, since a destroyed planting can be resown at the cost of seed and labor rather than losing the season outright. Land preparation and out-of-season are not susceptible, since a swarm arriving before a crop is planted or after it is harvested has nothing standing to destroy.[6] A cluster in a livelihood zone with no crop calendar has $\mathcal{S}(c, \delta) = \varnothing$ on every date and is never susceptible. Damaging-form arrivals excluded by this step are the same insects in the same places at an agronomically irrelevant time, and are carried through as a second falsification group.

The transcribed FEWS NET calendar fixes a normative planting date for each zone, but the rains that actually trigger planting arrive earlier or later than that norm in any given year and cluster, so the calendar is shifted cluster by cluster to track the rain rather than left pinned to its transcribed date. Season onset is detected separately for each cluster, year, and season from dekadal (one dekad is equal to ten days) CHIRPS rainfall, searched within a season-specific window, dekads 4 to 18 (roughly early February through late June) for the long rains and dekads 25 to 36 (September through December) for the short rains. Let ρ_d denote a cluster's rainfall total in dekad d of that window. Onset is the first dekad d satisfying three conditions simultaneously:

1. a trigger, $\rho_d \geq 25$ mm;

[6] Counting only the standing crop (rank 3) as susceptible raises the weight-for-height estimate to -0.215 at 30 km and -0.390 at 10 km, significant at the 5 and 1 percent levels respectively (Table 6).

2. a confirmation, that the one or two dekads immediately following together clear $\sum \rho_{d+1}^{d+2} \geq 20$ mm, so a single wet dekad followed by drought does not count;

3. and a dry-spell check, that at most one of those following dekads itself falls below 5 mm, so a trigger immediately undercut by a dry spell does not count either.

If no dekad in the window satisfies all three, onset is not detected for that cluster-year-season.

A cluster's climatological onset is the median onset dekad across 1981-2018 for that cluster and season. The transcribed calendar boundaries are shifted, holding stage lengths fixed, by the gap between a given year's detected onset and that climatological median, rounded to the nearest dekad and capped at six dekads (60 days) in either direction so a single anomalous year cannot pull the calendar implausibly far from its transcribed position; a cluster-year-season where onset could not be detected is left unshifted at its climatological position rather than dropped. A season that begins three weeks late accordingly shifts that cluster's susceptible window three weeks later rather than leaving it pinned to the calendar date.[7]

4.1.3 Verified damage

The third condition is that vegetation actually suffer. NDVI is observed in 16-day composites. Let $z_{c,t}$ be cluster c's NDVI anomaly z-score in composite t, and let t_ℓ index the composite containing event ℓ's observation date. Define the mean anomaly in the two composites preceding and the two composites following the event,

$$\bar{z}_{\text{pre}}(\ell) = \frac{1}{2}\sum_{k=1}^{2} z_{c(\ell),\, t_\ell - k}, \qquad \bar{z}_{\text{post}}(\ell) = \frac{1}{2}\sum_{k=1}^{2} z_{c(\ell),\, t_\ell + k}, \tag{6}$$

a window spanning roughly 16 to 48 days after the observation, depending on where within its own composite the event falls.[8] Composites with a missing value are dropped from the corresponding average rather than treated as zero. The change in anomaly is

$$\Delta z(\ell) = \bar{z}_{\text{post}}(\ell) - \bar{z}_{\text{pre}}(\ell), \tag{7}$$

and the event is classified as a verified decline if

$$\text{Decline}(\ell) = \mathbb{1}\big[\Delta z(\ell) \leq -\theta\big], \qquad \theta = 0.25, \tag{8}$$

fixed in advance. A locust record only counts as "verified damage" if the cluster's greenness fell by at least 0.25 standard deviations (θ). The threshold is not chosen from the health estimates. It only has to separate a fall in greenness from composite-to-composite noise, and the estimate barely moves with it: across thresholds from 0 to 1.5 and the six pre- and post-event windows, the weight-for-height estimate at 30 km stays between -0.134 and -0.213.[9]

[7] The shift matters at the wider radii. With the calendar left at its transcribed FEWS NET dates, the weight-for-height estimate at 30 km is -0.107 ($p = 0.16$), and widening every stage window by 15 days on each side gives -0.095 ($p = 0.18$). Narrowing every window by 15 days gives -0.168, close to the main estimate but imprecise ($p = 0.13$) on 911 treated children. Both looser calendars admit arrivals that more likely fell on bare ground. These are timing errors of the kind the stage condition exists to remove, and they water the estimate down toward the proximity null in the same way dropping the stage condition does (Figure 4). At 10 km the unshifted, widened and standing-crop variants remain significant at the 1 percent level and the narrowed variant at the 10 percent level (Table 6). The six-dekad cap binds in 19 of the 26,208 cluster-season-years from 2019 to 2022, under 0.1 percent.

[8] Using one, two or three composites on either side, in the six combinations in Table 6, gives weight-for-height estimates at 30 km between -0.134 and -0.211, all significant at the 5 percent level.

[9] For θ between 0 and 1 the weight-for-height estimate at 30 km lies between -0.177 and -0.212, significant at the 5 percent level or better throughout. At $\theta = 1.5$, which leaves 357 treated children, it is -0.213, significant at the 1 percent level (Table 6).

4.1.4 The exposure indicator

Treatment requires all three conditions to hold on a single record,

$$T_i^{(r)} = \mathbb{1}\Big[\exists\, \ell \in \mathcal{E}_i(r) : \text{Form}(\ell) = \text{damaging},\ \text{Susceptible}\big(S(c(i), \delta_\ell)\big) = 1,\ \text{Decline}(\ell) = 1\Big]. \tag{9}$$

All three conditions must hold on the same observation. A swarm in March and a vegetation decline in September are not evidence that the swarm caused the decline. Equation (9) admits only arrivals followed by a fall in the greenness of the same cluster within roughly six weeks. Each condition is set by how a swarm damages a crop rather than tuned to the outcome: form by the density at which locusts strip a field, stage by whether a crop stands in the field to be eaten, and decline by whether the cluster's vegetation then fell.

Equation (8) uses a variable closer to the outcome than the other two conditions, so three safeguards apply. The test in equation (7) is a change in anomaly rather than a level, so a cluster that was already dry does not qualify on dryness alone. The window in equation (6) is short enough that a later seasonal decline unrelated to the event is not attributed to it. Equations (6) to (8) are applied identically to benign records and to damaging-form records arriving outside the susceptible window. If Decline(ℓ) were capturing drought rather than locust damage, these placebo groups would satisfy it at a similar rate and predict outcomes similarly. Section 5.3 reports that they do not.

Three nested measures are defined by dropping conditions from equation (9). Proximity, $T_i^{(1,r)} = \mathbb{1}[\exists \ell \in \mathcal{E}_i(r) : P_\ell = 1]$, is gated on nothing beyond presence and is the closest analogue to the exposure variables used in existing locust research. Damaging form adds equation (1). Damaging form and stage adds equation (5). Then, by construction $T_i^{(r)} \le T_i^{(3,r)} \le T_i^{(2,r)} \le T_i^{(1,r)}$ and $T_i^{(r)} \subseteq T_i^{(3,r)} \subseteq T_i^{(2,r)} \subseteq T_i^{(1,r)}$.

My measure and the unverified proximity measures treat systematically different children as exposed.[10]

Table 1 compares the disputed children to those both measures agree were exposed. The main differences point to disadvantage among the disputed children. Disputed children have mothers with 3.5 fewer years of education, live 7.1 km farther from a major road, sit 0.66 of a wealth quintile lower, are 19 percentage points less likely to own land, and experienced gestational conditions 0.44 standard deviations drier on SPEI. Each difference is significant at the 0.1 percent level.

The difference comes mainly from livelihood zones. Among children both measures agree were exposed, 83 percent live in cropping zones, 15 percent in agropastoral zones, and none in pastoral zones. Among disputed children, 35 percent live in cropping zones and 43 percent in pastoral zones. By construction, a pastoral zone has no crop calendar, so no arrival there can satisfy the stage condition, and no child there can be treated. Proximity flags pastoral children at 64 percent at the 30 km radius, higher than the 48 percent rate in cropping zones, because the swarms arrived in the north and northeast first and stayed longest.

Disputed children have a mean weight-for-height of -0.47, below the -0.09 of children both measures agree were exposed and the -0.10 of unexposed children. The raw means show that the areas proximity adds are drier, poorer, and farther from a road.

[10] Table 9 and Figure 9 report counts at every radius.

Table 1: Characteristics of disputed children, 30 km radius

	Agreed	Disputed	Difference	p
Mother's education (years)	9.42	5.88	−3.54	< 0.001
Wealth quintile	3.00	2.35	−0.66	< 0.001
Distance to major road (km)	3.75	10.88	7.13	< 0.001
Owns land	0.545	0.359	−0.186	< 0.001
Owns livestock	0.662	0.672	0.010	0.506
Rural	0.640	0.646	0.006	0.715
Gestational SPEI (6 month)	1.265	0.827	−0.439	< 0.001
Gestational NDVI z-score	0.747	0.840	0.092	< 0.001
Livelihood zone shares				
Cropping	0.831	0.352	−0.480	
Pastoral	0.000	0.426	0.426	
Agropastoral	0.145	0.169	0.024	
Riverine	0.020	0.038	0.018	
Fishing	0.003	0.015	0.011	

Agreed denotes children flagged by both the proximity and verified measures (N = 1,760). Disputed denotes children flagged by proximity only (N = 2,310). Sample is the 7,603 children with valid anthropometry. p-values from two-sample t-tests.

4.2 Health Outcomes

The DHS measures every sampled child's length or height and weight under a standardized protocol: recumbent length for children under 24 months, standing height above that age, and weight to the nearest 100 grams, up to 59 months inclusive. DHS converts these raw measurements to three z-scores centrally, before the data are released, against the WHO Child Growth Standards, the international reference built from healthy, breastfed children raised without growth constraints (WHO Multicentre Growth Reference Study Group, 2006). The conversion uses the WHO's LMS method: at each value of the conditioning variable (age for HAZ and WAZ, height or length for WHZ) the reference population's distribution is summarized by three sex-specific parameters, a median M, a coefficient of variation S, and a Box-Cox skewness power L, and a child's raw measurement X is converted to a z-score

$$Z = \begin{cases} \dfrac{(X/M)^L - 1}{L\,S} & L \neq 0 \\ \dfrac{\ln(X/M)}{S} & L = 0, \end{cases} \tag{10}$$

so that $Z = 0$ is the reference median and $Z = -2$ is two standard deviations below it, in the reference distribution's own units, regardless of the child's age. I take DHS's own pre-computed values directly rather than recomputing them, and set to missing any value the WHO flags as biologically implausible: height-for-age below −6 or above 6, weight-for-height below −5 or above 5, and weight-for-age below −6 or above 5.

Weight-for-height (WHZ) compares a child's weight to the reference for a child of the same sex and the same attained height or length, with age entering only through which reference curve applies, so it nets out the height a child has already reached and asks only whether her current body mass is proportionate to it. This makes it responsive on a short horizon. Since it carries no growth history of its own, WHZ is sensitive to recent nutritional deficits, and the effect of a shock on it decays over the first two years rather

than accumulating the way a height deficit does. The clinical label for WHZ below -2 is *wasted*, and a child below -3 is severely wasted.

Height-for-age (HAZ) and weight-for-age (WAZ) are constructed identically, applying the same LMS transformation in equation (10) against the sex-specific WHO reference curves for length-or-height-for-age and weight-for-age respectively, in place of the weight-for-height curve, with the same WHO flags. They differ from WHZ, and from each other, only in which anthropometric ratio and which reference curve they use. HAZ compares height to the reference for a child's age rather than a child's height, so it is a cumulative record: a child's height on any given day reflects every month of growth, or growth faltering, since conception, and a single bad month barely moves it. A low HAZ score is a proxy of sustained undernutrition, repeated illness, or a poor environment operating over a long horizon, and the clinical label for HAZ below -2 is *stunted*. WAZ compares weight to the reference for a child's age; because age enters through both the numerator's growth history and the denominator's reference curve, WAZ is a composite of the other two, a child can have a low WAZ score because she is short for her age, because she is thin for her height, or both, and the score alone does not distinguish them. The clinical label for WAZ below -2 is *underweight*.

4.3 Estimation

I estimate

$$y_{ict} = \beta\, T_i^{(r)} + X_{ic}'\gamma + \alpha_c + \delta_t + \theta_{k(c)}\cdot t + \varepsilon_{ict}, \tag{11}$$

where y_{ict} is the weight-for-height z-score of child i in survey cluster c born in year-month t, $T_i^{(r)}$ is the exposure indicator at radius r defined in Section 4.1, α_c is a survey-cluster fixed effect, δ_t is a birth-cohort fixed effect defined at the year-month level, so that May 2020 and May 2021 occupy separate cells rather than being pooled as the same calendar month, and $\theta_{k(c)}\cdot t$ is a county-specific linear trend in that same year-month index.

The identifying variation is within cluster and across birth cohorts. Two children in the same enumeration area, born eight months apart, face different gestational windows. One window overlaps a verified damage event within r kilometers and the other does not. Equation (11) compares them.

I include survey cluster and birth-cohort (month×year) fixed effects. The county-specific linear trend allows counties to sit on different trajectories over the period. The main specification, column (3) of Table 2, which every other table follows, omits the trend; column (4) adds it.

The control set follows Conte et al. (2023). Child controls are sex, birth order, first-born status, twin status, and the interval since the preceding birth.

Family controls are mother's education, mother's height, household head age, household head sex, wealth quintile indicators, and the six-month SPEI over the gestational window. Mother's height enters because it captures the mother's own early-life nutrition and genetic factors that are strong predictors of a child's growth potential.[11] Wealth quintile indicators enter as a factor rather than a linear term because the relationship with anthropometry is not monotone in the data. Rainfall that is favorable enough to support three generations of locust breeding is also favorable for crops and pasture, so drought conditions are correlated

[11] Mother's height is recorded only for women in the subsample of households where DHS measured women, so it is missing for 48 percent of children with valid anthropometry. I fill a missing value with the median among measured women. Table 29 in Appendix A re-estimates all three outcomes adding an indicator for a filled value, dropping mother's height, and restricting to children whose mothers were measured (3,513 in the weight-for-height estimation sample). At 30 km the weight-for-height estimate is -0.184, -0.184 and -0.174 under the three alternatives, all significant at the 5 percent level or better. The weight-for-age estimate is -0.167 and -0.157 under the first two and falls to -0.112, no longer significant, among measured mothers. At 30 km height-for-age stays insignificant under every alternative.

with both the treatment and the outcome. I include SPEI over the child's own gestational window to remove the part of the estimate that reflects climatic water balance rather than the insects.[12]

Standard errors are clustered on a half-degree spatial grid, which yields 166 cells across the sample. These are coarser than the survey cluster. A swarm that arrives at one enumeration area almost always arrives at its neighbors, and the downstream effects on vegetation, local food availability, and prices propagate over a similar area. Residuals are therefore correlated across nearby clusters and not only within them, and clustering at the DHS cluster would overstate precision. The half-degree grid gives a wider-than-cluster but still local unit. Clustering at the county instead would leave 47 groups, too few for reliable asymptotic inference and too coarse to absorb variation.

All regressions are weighted by the DHS women's individual sampling weight (v005, divided by 1,000,000). The 2022 DHS is a stratified two-stage sample: enumeration areas are drawn with probability proportional to size within each stratum and a fixed number of households is drawn in each, and some areas are sampled at higher rates so that they yield precise estimates of their own. Households therefore enter with unequal probabilities. The weight is the inverse of the selection probability, adjusted for nonresponse in the stratum, and without it estimates lean toward the strata sampled at higher rates.[13] The weighted estimate is therefore the effect for Kenyan children born in this window rather than for the sample as drawn.[14]

5 Results

5.1 Health Outcomes and Radii

Table 2 reports the wasting estimates. Columns (1) to (4) use the weight-for-height z-score. Column (1) includes only the fixed effects, column (2) adds the child controls, column (3) adds the family controls, and column (4) adds county-specific linear time trends. Columns (5) and (6) repeat columns (3) and (4) with an indicator for wasting, a weight-for-height score below -2, as the outcome.

In utero exposure to a verified damage event within 30 km lowers weight-for-height by 0.207 standard deviations in the most conservative specification. This corresponds to a 3.6 percentage point rise in the probability of wasting in column (6), a 50 percent increase over the baseline of 7.2 percent among unexposed children. Without the county trend the estimate is smaller on both measures, 0.185 standard deviations on weight-for-height and 1.9 percentage points on wasting, a relative increase of 26 percent. The weight-for-height z-score estimate moves by less than 0.02 standard deviations as the child and family controls enter. The standard error is stable near 0.07 throughout and the weight-for-height z-score estimate is significant at the 1 percent level in every column.

Among children verified-exposed at the main radius, the number of confirmed damage events during gestation predicts weight-for-height beyond a simple binary indicator of exposure. Table 3 reports the relationship under three functional forms: the raw count of verified damage events entering linearly, the log of one plus that count, and the count capped at 20 to limit the influence of the small number of children exposed to an unusually large number of events. Each additional verified event is associated with a 0.013 standard deviation reduction in weight-for-height, significant at the 10 percent level; the log and capped forms

[12] Occupation and food consumption are excluded from the control set. These variables are unfortunately noisy self-reports in the DHS and are largely missing or the timing requested is irrelevant to the study; one week in most cases.

[13] See the DHS Program's Guide to DHS Statistics, which also notes that DHS no longer advises against weighting regressions.

[14] The weight distribution is slightly right-skewed. Appendix B describes it and reports the estimates under winsorized weights and without weights (Table 37). At 30 km the weight-for-height estimate is -0.164 with winsorized weights and -0.103 unweighted, both significant at the 5 percent level. The unweighted estimate is smaller, so the effect is larger among children who carry larger weights, those in strata the design sampled at lower rates.

Table 2: In utero locust exposure and child wasting, 30 km radius

	Weight-for-height z-score				Wasted	
	(1)	(2)	(3)	(4)	(5)	(6)
In utero exposure	−0.192***	−0.195***	−0.185***	−0.207***	0.019**	0.036***
	(0.072)	(0.070)	(0.069)	(0.074)	(0.008)	(0.009)
Child controls	No	Yes	Yes	Yes	Yes	Yes
Family controls	No	No	Yes	Yes	Yes	Yes
County time trend	No	No	No	Yes	No	Yes
Observations	7,443	7,443	7,416	7,416	7,416	7,416
R^2	0.344	0.351	0.352	0.361	0.261	0.270

All regressions include birth cohort and enumeration area fixed effects. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All regressions weight by the DHS individual sampling weight. Child controls are sex, birth order, first-born status, twin status, and the preceding birth interval. Family controls are mother's education, mother's height, household head age and sex, wealth-quintile indicators, and six-month gestational SPEI. Control coefficients are estimated but not reported. County time trend denotes county-specific linear trends in calendar month of birth. The outcome in columns (5) and (6) is an indicator for a weight-for-height z-score below −2, with a baseline prevalence of 7.2 percent among unexposed children. There are 1,703 treated children in the estimation sample.

return larger, more precisely estimated reductions of 0.237 and 0.029 standard deviations, both significant at the 1 percent level. All three specifications share the control set and fixed effects of column (3) of Table 2 and additionally control for the number of RAMSES ground visits recorded within 30 km during gestation, which separates the effect of confirmed damage from the effect of monitoring intensity in the same area during the same period.

Table 3: Dose response among verified-exposed children: number of damage events and weight-for-height

Dependent variable: Weight-for-height z-score	Linear	Log	Capped at 20
Verified damage events	−0.013*		
	(0.007)		
Log(1 + verified damage events)		−0.237***	
		(0.079)	
Verified damage events, capped at 20			−0.029***
			(0.010)
Observations	1,459	1,459	1,459

Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Sample restricted to children verified-exposed at the main 30 km radius; of the 1,757 verified-exposed children with complete data, those who are the only such child in their survey cluster or birth cohort drop out of the fixed-effects estimation, which leaves 1,459. All columns include the control set and the survey-cluster and birth-cohort fixed effects of column (3) of Table 2. I control for the number of RAMSES ground visits during gestation, counting every recorded visit within 30 km, of any outcome, during the child's gestation window, to separate the effect of confirmed damage from the effect of monitoring intensity in that area during that period.

Table 10 and Table 11 in Appendix A report the same three functional forms for height-for-age and weight-for-age, using the identical verified-exposed sample, control set, and ground-visit control. Weight-for-age moves in the same direction as weight-for-height: an additional verified damage event is associated with a 0.016 standard deviation reduction, and the log and capped forms return reductions of 0.279 and 0.037 standard deviations, all three significant at the 1 percent level. Height-for-age also falls with the number of events. The log and capped forms return reductions of 0.181 and 0.025 standard deviations, significant at the 5 percent level, and the linear form 0.011, significant at the 10 percent level.

I estimate the same specifications on the other two standard z-scores and their clinical thresholds. Height-for-age effects are negative at every radius but significant only at 20 and 25 km at the 5 percent level and at 10 and 15 km at the 10 percent level, so the height result depends on the choice of radius. At 30 km, the point estimate is −0.059 with a standard error of 0.060, and the probability of stunting is unchanged. Weight-for-age falls 0.167 standard deviations, though the probability of being underweight does not change (See Table 12).

Weight-for-height records a deficit of body mass relative to attained height, and it is an indicator of acute malnutrition. Height-for-age records cumulative linear growth from conception onward and responds only slowly, over quarters and years. The invasion ran for sixteen months, from late December 2019 until April 2021, but the median child with valid anthropometry was 14 months old at measurement and the mean 14.7 months. For the average child, therefore, the shock occupies the recent portion of a short life (WHO Multicentre Growth Reference Study Group, 2006).

Splitting the sample by age at measurement shows the weight-for-height effect declining with age (Table 13 in Appendix A). At 30 km the estimate is −0.410 for children measured before 12 months and −0.200 at 12 to 23 months, both significant at the 5 percent level, and 0.122 and insignificant at 24 months or older, where only 174 children are treated. Weight-for-age follows the same pattern, −0.400 before 12 months, significant at the 5 percent level, and −0.170 at 12 to 23 months, not significant. Height-for-age is insignificant before 24 months and −0.609 at 24 months or older, significant at the 5 percent level. The effect of in utero exposure on weight-for-height is therefore largest in the first year of life and decays over the first two years, which is why it is still visible at a median age of 14 months.

Weight-for-age is a composite of body mass and linear growth against age, so an acute weight deficit registers there but is diluted by the height component that has not moved. The continuous point estimate on weight-for-age is 90 percent of the continuous weight-for-height estimate, and it does not cross the underweight threshold.

Table 14 and Table 15 in Appendix A report the full stepped-control specification for height-for-age and weight-for-age in the same format as Table 2.

I also estimate the three health outcomes at all nine radii (Table 16 in Appendix A). Figure 3 plots the decay with confidence bands. Weight-for-height is significant at the 10 percent level or better at every one, and at the 5 percent level or better at every radius up to 30 km and at 45 km. The estimate is largest within 10 km, at 0.313 standard deviations, falls monotonically until 40 km (−0.135), ticks up at 45 km (−0.141), and is −0.119 at 50 km. The number of treated children rises from 698 to 2,210 over the same range. Widening the buffer adds children who were farther from any verified damage event and dilutes the average effect, and the estimate shrinks gradually.

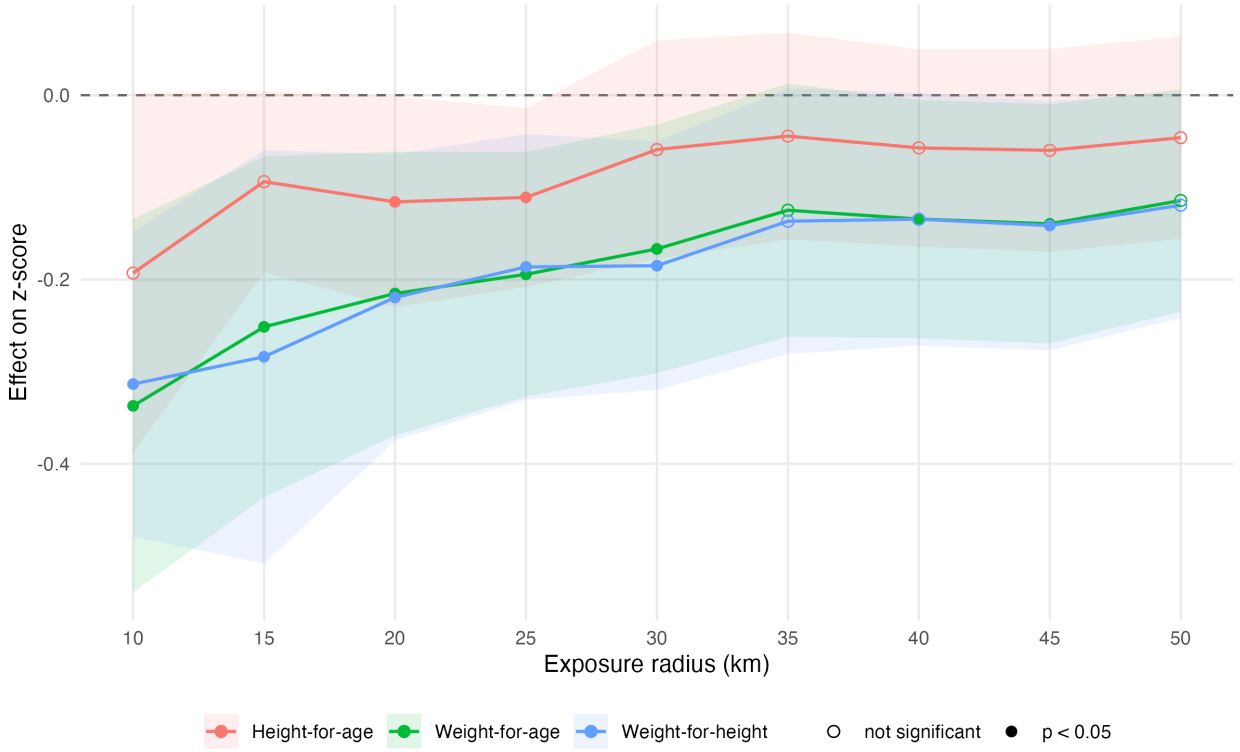


Figure 3: Estimate against radius, by outcome, with 95 percent confidence bands.

5.2 The Exposure Conditions

Figure 4 shows the three conditions of Section 4.1 each regressed at the main radius. Proximity alone has no detectable effect. Its weight-for-height estimate is −0.003 with a standard error of 0.056, and it flags 4,070 of the 7,603 children with valid anthropometry. Restricting to damaging form removes 79 children and moves the estimate to −0.035. Adding the crop-stage condition removes 1,263 more and moves it to −0.064, still insignificant. Adding the verification condition removes a further 968 and moves the estimate to −0.185, significant at the 1 percent level.

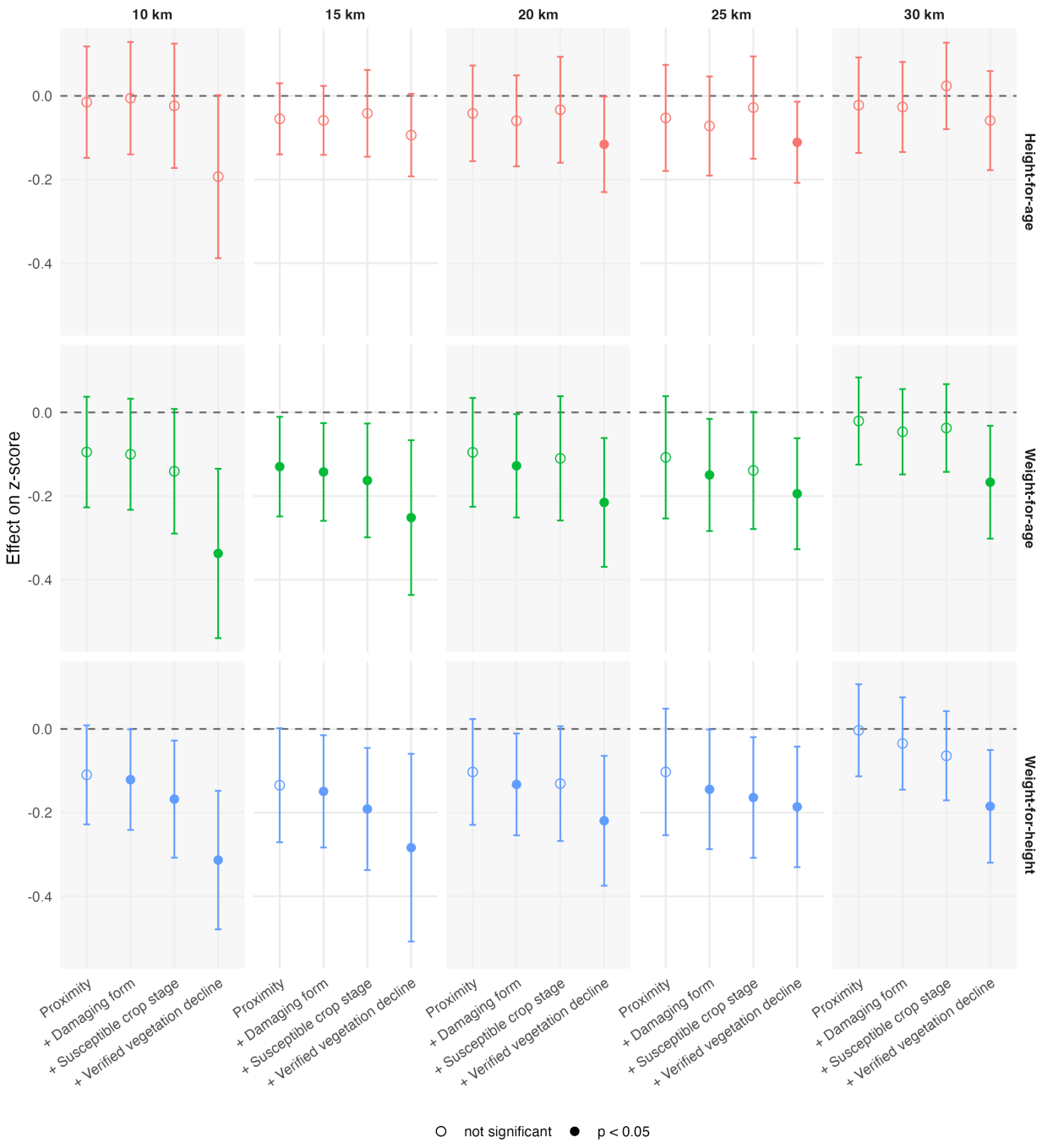


Figure 4: Point estimates with 95 percent confidence intervals.

5.3 Robustness

5.3.1 Falsification Tests

I provide three falsification tests for the exposure conditions. The first replaces damaging-form records with solitarious and transiens records and applies the stage and NDVI conditions unchanged. These are locusts at densities incapable of stripping a field. The second uses damaging-form records that arrive outside the

susceptible window, the same insects in the same places when there is nothing standing to destroy.

Applied naively, the solitary placebo flags children who are also flagged by the real treatment, because a cluster visited by solitary locusts during a gestational window can be visited by swarms in the same window. At 30 km, 86 percent of the children the naive placebo flags are also treated. Both placebo groups are therefore purged of any overlap with the treatment before estimation.

Panel A of Table 17 reports the purged tests. Neither recovers a negative and significant effect on weight-for-height at any radius. With the stage and vegetation conditions applied, the solitarious placebo can be estimated from 20 km and is never negative and significant; at 30 km it is 0.168 (standard error 0.324). Without those conditions it is 0.036 (0.103). The out-of-season placebo estimate at 30 km is 0.085 (0.062). Across 25 purged placebo cells on weight-for-height, none is negative and significant. One cell is positive and significant at the 5 percent level, the conditioned solitarious placebo at 25 km (0.641), which rests on a small group after purging.

The first purging exercise sets the placebo indicator to zero for treated children but leaves them in the comparison group, and because treated children have lower weight-for-height, this pushes the placebo estimates upward. Panel B of Table 17 therefore adds the verified indicator at the same radius as a control, and Panel C drops verified-exposed children from the sample. The solitarious placebos remain null under both. No estimate is negative and significant at the 5 percent level on weight-for-height, and the conditioned placebo at 25 km is 0.591 in Panel B and 0.463, insignificant, in Panel C. The out-of-season placebo is unchanged up to 35 km but turns negative at wider radii once verified exposure is controlled for: -0.115 at 40 km and -0.099 at 45 km, significant at the 10 percent level, and -0.128 at 50 km, significant at the 5 percent level. Across the three outcomes, 7 of 27 out-of-season estimates in Panel B are negative and significant at the 5 percent level, six of which are at 40 to 50 km and mostly on height-for-age and weight-for-age. With verified-exposed children dropped instead, none is. The falsification therefore holds within 35 km. At wider radii, damaging swarms that arrived out of season are associated with somewhat worse outcomes once verified exposure is held fixed, which may reflect damage just beyond the radius, a regional food shortfall, or other consequences of the swarms' presence.

Permuting the treatment indicator within survey clusters, 1,000 times, places the 10 km estimate on weight-for-height outside the entire simulated null with a randomization p-value below 0.001. The simulated standard deviation of the permuted estimate is 0.081, close to the clustered analytic standard error of 0.084. Running the same procedure on the solitarious placebo at 10 km, with the stage and vegetation conditions applied and before purging (-0.298), returns a randomization p-value of 0.482.

Third, I test whether any loss of greenness during gestation predicts worse outcomes, whether or not locusts were present. I treat every 16-day composite in a child's own cluster as a candidate event and apply the decline condition of equation (8) and the crop-stage condition of equation (5), without requiring a locust record. The 2014 round gives a direct version of this test, since no vegetation decline in the 2014 cohort's gestation windows can be locust damage. Declines that meet both conditions are common: 81 percent of 2014 children and 78 percent of 2022 children have at least one during gestation. I therefore also use two other versions: a matched version, which treats the 23 percent of children in each round with the largest decline, the same share as the verified measure at 30 km, and a count of qualifying declines. In 2014, none of the twelve estimates across the three versions and four outcomes is significant at the 5 percent level, and two are significant at the 10 percent level: any decline on height-for-age (-0.270) and the count on death (-0.004) (Table 18 in Appendix A). The matched estimate on weight-for-height is 0.005 with a standard error of 0.041.

In 2022 the matched decline does predict lower weight-for-height (−0.161, standard error 0.069), but only where locusts were recorded. Children with a matched decline and a locust record of any form within 50 km during gestation have weight-for-height 0.217 standard deviations lower, while children with a matched decline and no record nearby show no difference (−0.034, standard error 0.073; Table 19 in Appendix A). Adding the matched decline to the main specification leaves the verified estimate at −0.184.[15] A cluster with no locust record is not necessarily a cluster without locusts, so the 2022 comparison is less clean than the 2014 one. In neither round, however, does a loss of greenness away from recorded locust activity predict lower weight-for-height.

5.3.2 Drought

A natural threat to identification is a shock to vegetation that coincides with the invasion rather than following from it. The gestational SPEI control absorbs a smooth, cluster-specific measure of the climatic water balance, but it may not absorb all of it, and if verified exposure is systematically correlated with drought conditions that six-month index leaves unmeasured, the wasting estimate could partly reflect a drought effect misattributed to locusts.

Unconditionally, exposed children come from locations with a somewhat higher six-month SPEI on average than unexposed children, 1.265 versus 0.997 (Table 21 in Appendix A). This reflects where verified locust damage happened to occur rather than climatic differences caused by exposure. The gap reverses when the main specification compares children within the same cluster and birth cohort. Exposed children experienced somewhat drier gestational windows than their unexposed neighbors in the same cluster and cohort, −0.18 standard deviations on the six-month SPEI index, although the gap is not statistically significant ($p = 0.13$). This within-cluster gap runs in the direction that would make a drought confound plausible in principle. If drought lowers weight-for-height, a treated group with drier gestations would push the exposure coefficient in the negative direction this paper reports.

I remove the SPEI variable and re-estimate at every radius (see Table 22 in Appendix A). Since treated children had drier gestations within cluster, dropping SPEI should make the coefficient more negative if drought matters, and it does at every radius. The change is 0.005 to 0.008 standard deviations, about a tenth of a standard error.

Additionally, NDVI and SPEI are not interchangeable measures of the same underlying conditions. SPEI is a climatic water-balance index, precipitation minus potential evapotranspiration, standardized against a location's own history. It describes moisture stress acting on every hectare in a cluster roughly uniformly, whether or not a crop or pasture stands on it. NDVI records the greenness of the vegetation actually present on the ground, and it responds to anything that removes standing biomass abruptly, not only to a shortfall in the water available to grow it. A locust swarm defoliating a field over a matter of days produces a sharp, localized drop in NDVI on a timescale a slow-moving drought's drawdown on the water balance does not resemble. Equation (8) is designed to detect a change over roughly six weeks around a specific arrival rather than a seasonal level. This rests on the assumption that no other large-scale consumer of vegetation, a wildlife grazing episode or an unrelated pest outbreak, stripped the same cluster in the same narrow window

[15] In 2022 the count of qualifying declines is also associated with lower height-for-age (−0.038) and weight-for-age (−0.036), both significant at the 1 percent level (Table 18). Split by whether a locust record fell within 50 km (Table 20), declines near locust activity predict lower height-for-age (−0.039) and weight-for-age (−0.040), both significant at the 1 percent level, and lower weight-for-height (−0.027, significant at the 10 percent level). Declines away from locust activity also predict lower height-for-age (−0.035, 10 percent level) and weight-for-age (−0.029, 5 percent level), but not weight-for-height (−0.016, standard error 0.016). The count therefore picks up some vegetation stress beyond locust damage on the slower-moving outcomes, while the weight-for-height result is confined to declines near recorded locust activity.

as a documented locust arrival. Kenya's rangelands do carry wild herbivores, but a defoliation event that happens to coincide with a verified locust record's own cluster and six-week window, systematically enough to generate the pattern in Section 5, would have to track the invasion's own arrival dates and locations rather than occur at random with respect to them.

Equation (8) tests a change in NDVI anomaly over roughly six weeks around a specific locust arrival, not a level of dryness over the season, so a cluster that is dry for the whole gestational window does not qualify on that dryness alone. Both placebo groups sit in the same drought-affected places and the same drought years as the real treatment, so if the decline condition were capturing an ambient dry spell rather than an event-specific loss of vegetation, they should satisfy it and predict the outcome at a similar rate. The test holds the climatic background fixed and asks whether a specific arrival was followed by a decline in vegetation, so it does not depend on the direction of the seasonal SPEI comparison. [16]

5.3.3 Mortality

The main results condition on survival. A child who died before the survey has no anthropometric measurement, so the sample-construction filter that requires a valid z-score removes them. If the shock also raised mortality among the most severely affected children, the ones who would have been measured as wasted, the survivors are positively selected and the estimates are biased toward zero.

Additionally, the weaker evidence on an effect on stunting in my analysis might be an artifact of mortality. Stunting is a long-run outcome while wasting is not. Many children might not have survived to an age where the long-run measure of health is observed.

I rebuild the birth panel without the anthropometry filter and re-run the identical exposure construction against it. The 2022 round has 8,213 births under this definition. Of these, 7,603 have valid measurements, 263 (3.2 percent) did not survive to interview, and 347 survived but were not measured. The mortality regressions use 8,057 of the 8,213 births: 27 are missing household head age or gestational SPEI, and a further 129 are the only child in their cluster and drop out of the cluster fixed-effects estimation. Table 23 regresses survival on exposure with the same controls, weights, fixed effects, and clustering. Nothing is significant at any radius. Estimates run from -0.001 to 0.005 against a baseline mortality rate of 3.2 percent, and the smallest p-value across the nine radii is 0.37. The three looser measures return the same null at the main radius.

Mortality selection therefore does not explain the weight or height results, and the weight results do not extend to survival.

5.3.4 Pre-existing Conditions

If the exposed clusters were simply on a worse trajectory, a treatment constructed the same way in a pre-invasion period should reproduce the result. I test this using the 2014 DHS round, conducted six years before the invasion.

I shift the locust events back 93 months — the gap between the KDHS 2014 and KDHS 2022 fieldwork periods — landing the invasion's calendar pattern on the equivalent point in the 2014 cohort's actual gestation windows, and re-evaluate the stage and NDVI conditions at the shifted dates using rainfall and NDVI from that period. Event locations and 2014 birth timing are unchanged. Whether a 2014 child's gestational window overlaps the shifted pattern then varies within cluster as real exposure does.

[16] See Figure 10 in Appendix A for more detail.

Across four measures, nine radii, and four outcomes — mortality and the three anthropometric measures — 144 cells in total (Tables 24 to 27 in Appendix A), none is significant at the 5 percent level. Five (3.5 percent) are significant at the 10 percent level, fewer than chance alone would produce, and three of the five are negative mortality estimates under the verified measure.

5.4 Potential Mechanisms

5.4.1 Anticipation, Crop Failure and Prices

Conte et al. (2023) argue that a locust invasion reaches children through two channels. An anticipatory channel operates during the growing season, when local prices rise on expected crop loss before any loss is realized. A crop failure channel operates afterward, when the harvest actually fails. For Mali they report that children exposed only to the anticipatory channel are harmed about as much as children who also experienced crop failure, and that both channels show up in local crop prices, with an 8.4 percent price increase during the speculative window that fully accounts for the anticipatory health effect once controlled for.

The authors argue that in Mali, the whole country grows crops on one calendar, one planting season, one harvest, once a year. In this setting, when locusts hit in the growing season, there is a shared window where every affected family already knows the harvest is in trouble but the crop has not failed yet.

In Kenya, rain falls twice a year, with many growing seasons for different crops, and the exact timing differs across the 17 livelihood zones that grow crops. The invasion also lasted sixteen months across two separate waves. Because no single date can serve as the point of common knowledge, I construct an area-specific boundary date, set by the expected completion of that area's crop.

For a child to count as "anticipatory," they must be born after the first exposure event within the radius of their cluster, so the household knew about it, but before that event's crop was due to finish its harvest. A child conceived after that harvest boundary falls in the crop failure group, and a child whose gestation spans the boundary falls in both. As in Conte et al. (2023), the groups are defined by when a child's gestation falls relative to the area's own event and boundary rather than by the child-level indicator of equation (9). At 30 km the weight-for-height estimation sample holds 613 anticipatory children, 1,422 in both windows and 1,174 in the crop failure group.

Table 31 in Appendix A enters the three groups jointly, with the control set and fixed effects of column (3) of Table 2, alongside children in pastoral and fishing zones who had a locust record nearby. At 30 km anticipatory children have weight-for-height 0.297 standard deviations lower (standard error 0.110), significant at the 1 percent level. Children in both windows are 0.150 lower (0.096), not significant, and the crop failure group shows nothing (0.002, standard error 0.107). Across radii the anticipatory estimate is significant at the 5 percent level from 10 to 35 km and at the 10 percent level from 40 to 50 km, and it is largest close to the damage (-0.453 at 10 km). On height-for-age it is -0.156 at 30 km ($p = 0.11$) and significant at the 5 percent level from 10 to 25 km and at 35 km.

In Kenya the anticipatory window is short. Most verified events occur while the crop is standing in the field rather than at planting [17], so the harvest boundary follows the arrival closely, and anticipatory children were mostly in the last months of gestation when the swarm destroyed the standing crop. Conte et al. (2023) identify an anticipatory channel by showing that local prices account for it. Adding the gestational maize price index here moves the 30 km estimate from -0.297 to -0.301.

[17] 72 percent of verified event-cluster pairs within 30 km fall in the standing-crop stage

I also test the price channel directly. Price data comes from the World Bank Real Time Prices series for Kenya, 233 markets with monthly maize and bean estimates from 2015 to 2022, of which 231 enter the regression. The test below uses each reporting market's own price series directly, with no interpolation to a DHS cluster or a county. A market is treated at radius r if a verified event ever fell within r kilometers of it, and its boundary is the harvest completion date of the earliest such event, the same construction used for the health regression above. Calendar time is measured for every market as quarters relative to that boundary rather than as a shared national quarter. A market with no verified event within radius is control throughout its own series. Maize and beans are pooled into one regression since the boundary comes from whichever crop made the governing event susceptible.

$$\log(\text{Price}_{m,t}) = \sum_j \theta_j \, T_m \times Q_j(t) + \mu_m + \delta_{s,t} + \xi_c \cdot t + \varepsilon_{m,t}, \tag{12}$$

where T_m is the market's own treatment status at radius r, $Q_j(t)$ is the market's own relative quarter, μ_m is a market fixed effect, $\delta_{s,t}$ is a staple by year month fixed effect, and $\xi_c \cdot t$ is a county specific linear trend. Standard errors are clustered on market.

Figure 5 plots $\hat{\theta}_j$ at the main 30 km radius. Two and three quarters before the boundary, treated markets run 2.7 to 3.4 percent above control markets, both significant at the 5 percent level. The coefficient falls toward zero in the quarter before the boundary and the quarter of the boundary itself. From one quarter after the boundary the coefficient turns negative, and it is significant from two quarters after. Treated markets run 3.2 percent below control markets two quarters after and 4.1 percent below three quarters after, both significant at the 1 percent level.

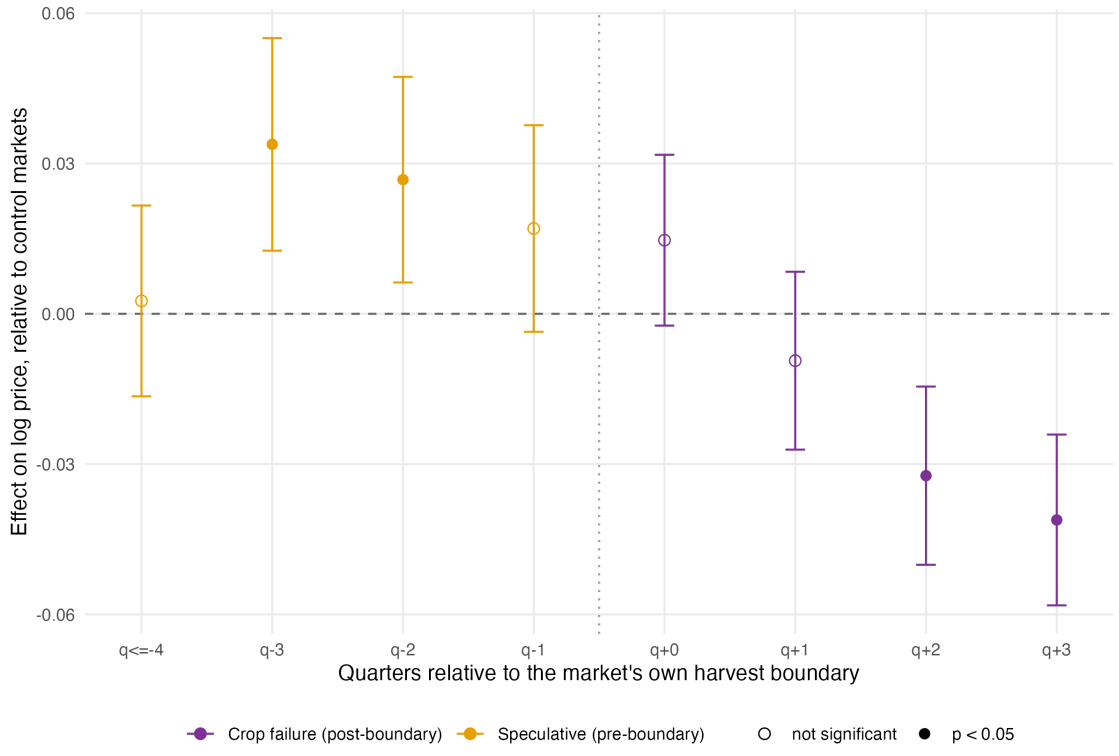


Figure 5: Point estimates with 95 percent confidence intervals. Market level treated versus control price event study. Quarters are relative to each treated market's own harvest boundary.

For the median treated market the harvest boundary falls one month after its first verified arrival, so most quarters before the boundary are also quarters before any swarm had been verified nearby. Figure 6 re-centers event time on each of the 83 treated markets' first verified arrival within 30 km. Treated markets already run 3.4, 3.1 and 2.7 percent above control markets three, two and one quarter before that arrival, all significant at the 5 percent level or better, and the gap narrows to 1.9 percent in the quarter of arrival. A premium that predates any news of the swarm is unlikely anticipation of crop loss. It is a pre-existing difference between the markets that verified damage later reached and the rest, and it shrinks once the swarms arrive.

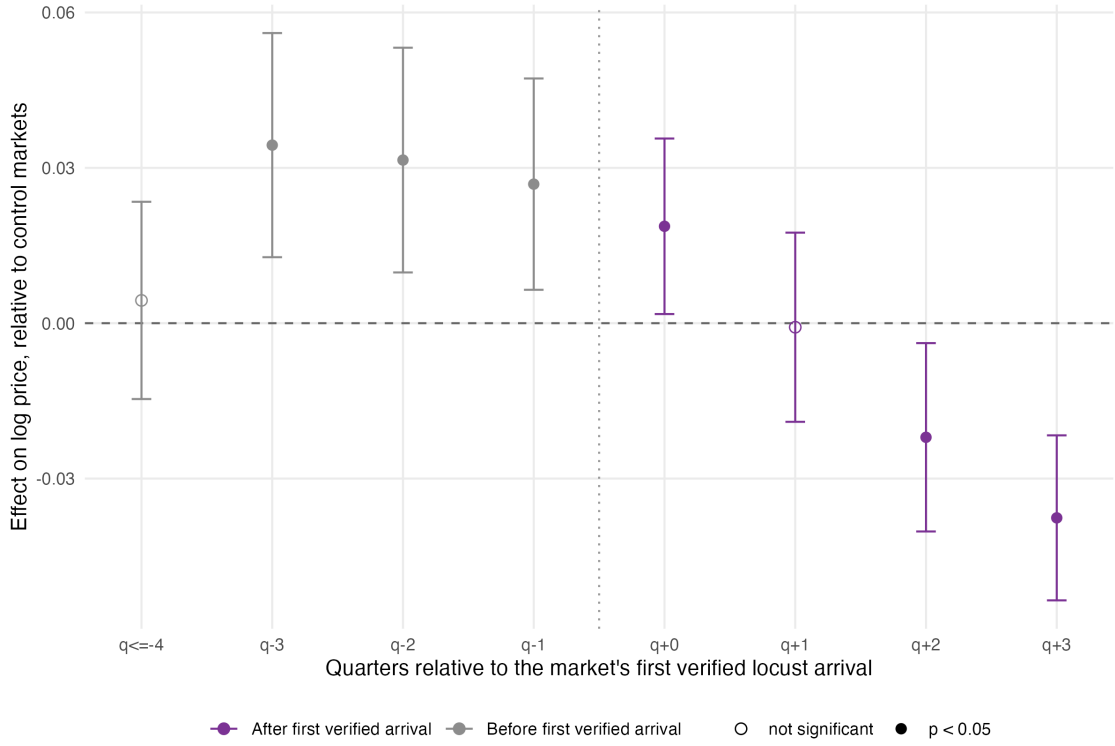


Figure 6: Point estimates with 95 percent confidence intervals. Market level treated versus control price event study at 30 km. Quarters are relative to each treated market's first verified locust arrival.

The pattern after the boundary does not fit a crop failure either. A failed harvest should raise the price of the crop that failed, yet treated markets fall below control markets from two quarters after the boundary. Together, Figures 5 and 6 are consistent with treated market pricing becoming dearer than control markets before the first verified arrival and then drifting back to, and slightly below, the control level over the following year, with no break at either the arrival or the harvest boundary. A price response to the swarms would show a break at one of those dates. The RTP series also fills most market-months by imputation from nearby markets and related commodities, which smooths local price movements and limits what it can detect.

The price data therefore support neither the anticipatory channel nor the crop failure channel. In Mali, Conte et al. (2023) find prices rising with the swarms and the price control absorbing the anticipatory health effect. Here the premium predates the first verified arrival, and controlling for prices moves the anticipatory estimate from -0.297 to -0.301. The anticipatory result then shows that the harm falls on children who

were in late gestation when the swarm destroyed the standing crop.[18]

5.4.2 Psychological Stress of Proximity

Figure 7 shows estimates at restricted samples - pastoral and fishing zones - and regresses each outcome on proximity exposure at every radius. None of the estimates is statistically significant. Weight-for-height estimates are positive at all nine radii, between 0.006 and 0.121, and weight-for-age estimates are positive at seven, on 642 to 1,203 treated children. Pastoral children are among the most often flagged by proximity, 64 percent at 30 km, yet I find no effect on their anthropometric outcomes.

This is consistent with two things. First, it is consistent with the absence of a general channel operating through the presence of locusts on mental health rather than through physical deprivation from hunger. It also suggests that the proximity measure's null at 30 km is not only a matter of power. The standard errors here are about twice those of the full-sample proximity estimate, but the point estimates on weight-for-height are positive at every radius rather than small and negative.

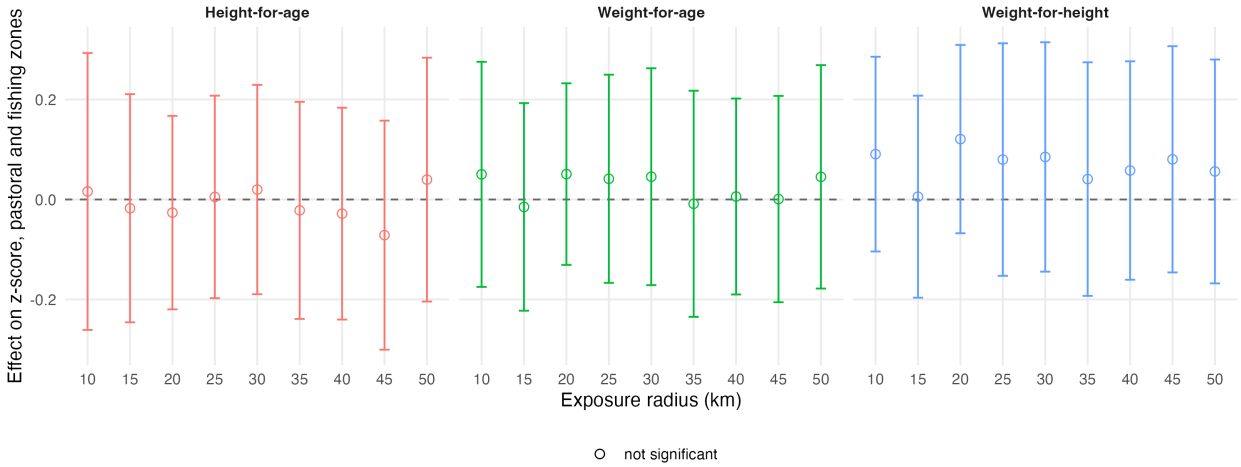


Figure 7: Proximity exposure in pastoral and fishing zones. Point estimates with 95 percent confidence intervals. No estimate is significant at the 10 percent level.

I run a similar test on a sample that was only treated when crops are not susceptible. Gregarious swarms of locusts that arrive outside a susceptible crop stage have no crop-loss mechanism at all, since nothing was standing in the field to threaten. These are "off-season" exposures. Run on the cropping zone sample (Table 30 in Appendix A), 1 of 27 estimates is significant at the 5 percent level, height-for-age at 50 km (-0.248, $p = 0.032$), and weight-for-height at 30 km is 0.240 ($p = 0.072$). This comparison group, however, includes children treated under the stage or verified measures, whose lower anthropometry pushes the off-season estimate upward. Controlling for both indicators at the same radius (Panel B), weight-for-height at 30 km falls to 0.205 and is insignificant, and 3 of 27 estimates are negative and significant at the 5 percent level, all at 40 and 50 km: height-for-age at 50 km (-0.281) and weight-for-age at 40 and 50 km (-0.300 and -0.359). Dropping those children instead (Panel C) leaves 1,667 to 3,393 children and no negative estimate significant at the 5 percent level, while weight-for-height at 30 km rises to 0.514 ($p = 0.035$). Within about 35 km, off-season exposure shows no consistent effect. At the widest radii it is associated with

[18] This matches the patterns in Section 5.5.2

lower height-for-age and weight-for-age when the stage and verified indicators are held fixed.[19]

5.5 Heterogeneity

5.5.1 Which Households Are Hit Hardest

Table 4 compares exposed children who are actually wasted to exposed children who are not, at the main 30 km radius. Wasted cases have mothers with 3.0 fewer years of education, sit 0.69 of a wealth quintile lower, and live 1.3 km farther from a major road. Underweight cases show a similar gradient, with mothers 1.2 cm shorter and a 13 percentage point higher rural share (Table 32 in Appendix A). Gestational SPEI is modestly lower among wasted cases (-0.17 standard deviations, $p = 0.118$) and gestational NDVI is modestly higher among underweight cases ($p = 0.015$), but both gaps are small next to the education, wealth, and distance gaps above, which are larger and more consistently significant. At the 15 km radius (Table 33 in Appendix A), neither gestational SPEI nor NDVI differs significantly between wasted and non-wasted children, and among underweight children only the NDVI gap approaches significance ($p = 0.057$).

Table 4: Exposed children who are wasted versus exposed children who are not, 30 km radius

	Wasted	Not wasted	Difference	p
Mother's education (years)	6.56	9.58	-3.02	< 0.001
Wealth quintile	2.34	3.04	-0.69	< 0.001
Distance to major road (km)	4.99	3.69	1.30	0.073
Rural	0.700	0.637	0.063	0.210
Owns land	0.444	0.551	-0.106	0.052
Mother's height (cm)	160.43	160.63	-0.20	0.591
Gestational SPEI (6 month)	1.100	1.274	-0.175	0.118
Gestational NDVI z-score	0.702	0.750	-0.048	0.571
Age at measurement (months)	14.58	15.73	-1.16	0.079

Sample restricted to children flagged as exposed at 30 km. Wasted denotes a weight-for-height z-score below -2 ($N = 90$). Not wasted denotes the remainder of the exposed group ($N = 1{,}669$). p-values from two-sample t-tests.

Table 5 reports these interactions, for all three anthropometric outcomes.

The top panel estimates, separately for each outcome $y \in \{\text{WHZ}, \text{HAZ}, \text{WAZ}\}$ and each candidate characteristic $Z_i \in \{$agricultural occupation, owns land, rural, distance to major road$\}$, twelve regressions of the form

$$y_{ict} = \beta\, T_i^{(r)} + \lambda\, Z_i + \eta\, \left(T_i^{(r)} \times Z_i\right) + X_{ic}'\gamma + \alpha_c + \delta_t + \varepsilon_{ict}, \tag{13}$$

one regression per (characteristic, outcome) pair, so the four rows in a column come from four distinct models, and the panel from twelve, rather than from one shared regression. $T_i^{(r)}$ is exposure at the main radius, and X_{ic}, α_c, and δ_t are the same family-and-child control vector, survey-cluster fixed effect, and birth-cohort fixed effect as column (3) of Table 2; the county-specific trend in equation (11) is not included here. The reported coefficient is η, the interaction between exposure and the characteristic; β and λ are not shown.

The bottom panel instead estimates one regression per outcome that includes every livelihood zone jointly. Let $D_{iz} = \mathbb{1}[\text{zone}_i = z]$ for each zone $z \in \{$pastoral, agropastoral, riverine, fishing$\}$, with cropping omitted

[19] A livelihood zone with no crop calendar can produce neither a crop-stage nor a verified damage record, so every damaging-form encounter a pastoral child has is biology-only by construction, and pastoral children are already poorer, more remote, and on drier gestational conditions than the rest of the sample.

as the reference zone:

$$y_{ict} = \beta\, T_i^{(r)} + \sum_z \lambda_z\, D_{iz} + \sum_z \eta_z \left(T_i^{(r)} \times D_{iz}\right) + X_{ic}'\gamma + \alpha_c + \delta_t + \varepsilon_{ict}, \tag{14}$$

with the same control set and fixed effects as equation (13). Cropping's own exposure effect is the coefficient β on $T_i^{(r)}$ alone, not reported in the table; each η_z is that zone's exposure effect relative to cropping's, so the three reported rows in a column all come from the same regression and are directly comparable to one another. No pastoral child is ever treated under the verified-damage measure, so $T_i^{(r)} \times D_{i,\text{pastoral}}$ has no variation to identify η_{pastoral} from and is dropped. Rural residence blunts the exposure effect significantly on weight-for-height and weight-for-age, an offset of 0.24 standard deviations on each, significant at the 5 percent level, though not on height-for-age; agricultural occupation moves the same direction on weight-for-height alone, at the 10 percent level. Land ownership is insignificant and distance to a road is significant at the 10 percent level for height-for-age and weight-for-age. These four characteristics are correlated with one another in this sample. Rural residence alone correlates −0.62 with wealth quintile and 0.39 with livestock ownership. They plausibly proxy a single underlying divide between subsistence-oriented and market-dependent households rather than four independent channels. Entering all four jointly in one regression, rather than one at a time (Table 34 in Appendix A), is consistent with this reading. No characteristic keeps its significance. Rural residence retains the largest point estimates, 0.206 on weight-for-height and 0.199 on weight-for-age, with standard errors about half again as large as in the one-at-a-time regressions. Rural households appear to fare better under exposure, but the data cannot distinguish whether land, livestock, occupation, or market access accounts for it.

The bottom panel of Table 5 interacts exposure with livelihood zone, with cropping zones as the reference category rather than each zone's own isolated treatment slope, since cropping holds 1,463 of the 1,760 treated children with valid anthropometry at the main radius and is the natural baseline for the comparison. Riverine zones fare worse than the cropping baseline on all three outcomes; an extra −0.25 standard deviations on weight-for-height ($p = 0.076$), −0.39 on height-for-age ($p = 0.034$), and −0.37 on weight-for-age ($p = 0.007$).

Riverine households are, if anything, less rural than households in the other zones, 31 percent against 60 to 76 percent, so the riverine gap is not a restatement of the rural result. It more plausibly reflects riverine households' dependence on flood-recession agriculture. Riverine households source roughly 55 percent of their food from their own production, against about 45 percent in cropping zones (Famine Early Warning Systems Network (FEWS NET), 2011). Re-estimated at every radius from 10 to 50 km (Table 35 in Appendix A), the riverine coefficient is negative and significant at the 10 percent level or better on all three outcomes at 30 and 35 km, on weight-for-height from 20 km, and on weight-for-age from 15 km, and it is indistinguishable from zero beyond 35 km.

Agropastoral zones show a statistically significant protective effect relative to cropping on all three outcomes, resting on 256 treated children across 80 clusters and 11 counties, though concentrated in the Western Agropastoral zone. Fishing zones return a large, significant positive coefficient on all three outcomes, but it rests on six treated children in a single grid cell in one fishing zone on Lake Victoria.

Table 5: Heterogeneity: 30km exposure interacted with baseline characteristics and livelihood zone

	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score
Panel A: Baseline characteristic			
Agricultural occupation	0.212*	0.046	0.167
	(0.116)	(0.125)	(0.128)
Owns land	0.095	0.006	0.067
	(0.102)	(0.072)	(0.102)
Rural	0.239**	0.112	0.239**
	(0.111)	(0.074)	(0.093)
Distance to major road (km)	0.011	0.018*	0.018*
	(0.010)	(0.009)	(0.010)
Panel B: Livelihood zone (relative to cropping)			
Agropastoral	0.254**	0.329**	0.339***
	(0.109)	(0.141)	(0.101)
Riverine	−0.246*	−0.390**	−0.369***
	(0.137)	(0.182)	(0.136)
Fishing	0.266***	0.609***	0.475***
	(0.094)	(0.103)	(0.093)

Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Panel A: each cell reports the coefficient on exposure (30km) interacted with the row characteristic from a separate regression of the column outcome on exposure, the row characteristic, their interaction, and the control set and fixed effects of column (3) of Table 2. Panel B: each cell reports exposure interacted with livelihood zone from a single regression per outcome, cropping as the reference level, with the same control set and fixed effects; each coefficient is that zone's exposure effect relative to cropping. Pastoral zones are excluded because no pastoral child is ever treated under the verified-damage measure. Among the 1,760 treated children with valid anthropometry at the main radius, 1,463 are in cropping zones, 256 in agropastoral, 35 in riverine, and 6 in fishing; the riverine and fishing coefficients rest on far fewer treated children than the cropping and agropastoral estimates.

5.5.2 Timing Within Gestation

Figure 8 splits verified exposure by the trimester in which the qualifying event falls, at every radius, on all three health outcomes.

First-trimester exposure is never significant on any outcome at any radius. The third-trimester estimate is largest close to the damage and falls with distance. On weight-for-height it is −0.368 at 15 km and −0.174 at 30 km, and from 35 km onward it lies between −0.050 and −0.078 and is insignificant. On weight-for-age it falls from −0.408 at 15 km to −0.211 at 30 km, and beyond 30 km it is at most significant at the 10 percent level (−0.110 at 40 km). The second-trimester estimate does not fall with distance. On weight-for-height it lies between −0.209 and −0.299 at every radius, is significant at the 1 percent level from 25 to 50 km and at the 5 percent level at 10 and 20 km, and only at the 10 percent level at 15 km. On weight-for-age it lies between −0.173 and −0.259 from 15 to 50 km and is significant at the 5 percent level or better from 20 to 50 km and at the 10 percent level at 15 km. At 10 km, where each trimester holds roughly 250 treated children, the second-trimester estimate on weight-for-age is not significant.

Which trimester carries the larger estimate therefore depends on the radius. The third trimester is larger at 15 and 20 km on weight-for-height and from 10 to 25 km on weight-for-age. The second trimester is larger at 10 km and from 25 km on weight-for-height, and from 30 km on weight-for-age. At 30 km the second-trimester estimate is −0.299 against −0.174 for the third trimester on weight-for-height, and −0.259 against −0.211 on weight-for-age. The pooled estimate in Table 16 averages over both.

The third-trimester component follows the same distance gradient as the pooled estimate in Section 5. It is large where verified damage was close to the household and shrinks as a wider radius adds children whose households were farther from it. A loss of the household's own crop during the months of fastest fetal weight gain would produce this pattern.

The second-trimester component does not depend on distance in this way. A child exposed in the second trimester is born roughly three to six months after the event, usually after the affected crop's harvest was due (Section 5.4.1), into whatever local food shortfall the lost harvest left, which is not confined to the households whose own fields were hit. A shortfall of this kind reaches beyond the immediate area of a swarm, which would explain why the second-trimester estimate on weight-for-height is about as large at 50 km (−0.209) as at 15 km (−0.234). This interpretation has three limits. The market-level price event studies in Section 5.4.1 show no price increase after the first verified arrival or the harvest boundary, so if the shortfall operated it would have to run through household stocks, local trade, or transfers the price data do not capture. Children conceived after the harvest boundary show no effect (Table 31), so any shortfall did not last long enough to reach the next cohort. The second-trimester children also sit mostly in the both group of that table, whose estimate (−0.150) is smaller and not significant. That group is defined from the cluster's first verified event only and can include children whose gestation began after the event, so it is not a clean counterpart of the trimester split, but the gap is a further reason for caution. Last, children born into a shortfall are also exposed to it in early infancy, so the second-trimester estimate may partly reflect postnatal rather than prenatal exposure.

Height-for-age follows the third-trimester pattern. The pooled estimate in Table 16 shows no significant effect on height-for-age at the main radius. Split by trimester, third-trimester exposure is significant on height-for-age from 10 to 30 km (−0.284 to −0.164, all at the 5 percent level or better), while first- and second-trimester exposure are never significant at any radius. The pooled null at 30 km is an average of a radius-limited third-trimester effect and two null trimesters. The third trimester is the period of fastest fetal linear growth, and Stein and Susser (1975) find a smaller but real effect of third-trimester famine exposure on length at birth alongside the larger effect on birth weight.

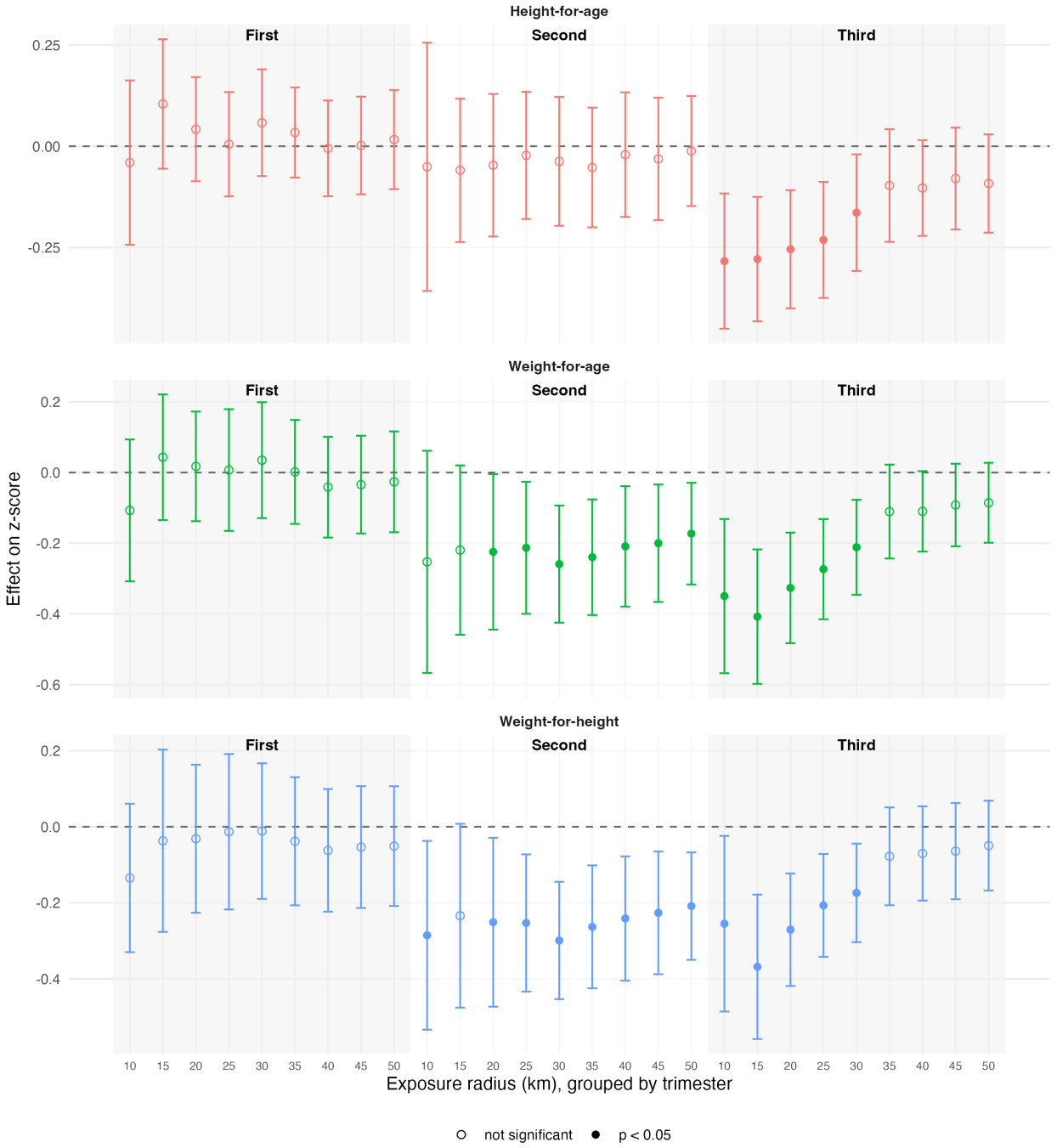


Figure 8: Regressions on an indicator for verified exposure occurring during that trimester, sharing the control set and fixed effects of column (3) of Table 2. Standard errors are clustered on a half-degree spatial grid.

5.5.3 Postnatal Exposure

The exposure measure is defined on the gestation window only. A child born in January 2020, for example, lived through the invasion as an infant but is included in the comparison group. If postnatal exposure also lowered weight-for-height, this would bias the estimate toward zero. I define postnatal exposure as a verified

damage event within 30 km of the child's cluster in the first 12 months after birth and before the interview. Among the 7,603 children with valid anthropometry, 1,128 meet this definition, and 772 of them are not exposed in utero. Adding it to the main specification moves the in utero estimate from −0.185 to −0.145 (standard error 0.079), significant at the 10 percent level, while postnatal exposure itself is 0.106 (standard error 0.072) and insignificant (Table 28 in Appendix A). Splitting postnatal exposure into the first and second half of the first year gives two positive estimates, neither significant at the 5 percent level, and among children not exposed in utero, postnatal exposure is 0.138 (standard error 0.082). The in utero estimate loses precision because the two indicators overlap: 356 of the 1,128 children exposed after birth were also exposed in utero, and its standard error rises from 0.069 to 0.079. Postnatal exposure itself is positive in every column, so harmed postnatally exposed children in the comparison group are not pushing the main estimate toward zero. Holding postnatal exposure fixed moves the in utero estimate toward zero instead, from −0.185 to −0.145, so the main estimate partly picks up the difference between children exposed before birth and children exposed only after it.

6 Discussion

Within 30 km of a verified damage event, in utero exposure lowered weight-for-height by 0.19 to 0.21 standard deviations and raised the probability of wasting by 1.9 to 3.6 percentage points against a baseline of 7.2 percent. The estimate is negative and significant at the 10 percent level or better at every radius from 10 to 50 km and is larger at smaller radii. It is concentrated in the second and third trimesters. The third-trimester effect is largest close to the damage and fades with distance, while the second-trimester effect is roughly constant across radii.

These results depend on how exposure is defined. A measure that records locust presence within 30 km of a household returns a coefficient of −0.003 on the same children, outcome, controls, and fixed effects. Requiring the record to describe a gregarious form and to arrive during a susceptible crop stage moves the estimate to −0.064, still indistinguishable from zero. Requiring a decline in vegetation over the following 16 to 48 days moves it to −0.185, significant at the 1 percent level. Just over half of the children the proximity measure flags at 30 km show no evidence of vegetation damage around them, and 43 percent of those live in pastoral zones where no crop calendar applies. A swarm that passes over bare ground or rangeland does not destroy a crop, and a measure that counts it as exposure dilutes the estimate toward zero.

The main limitation of this paper concerns the price channel. Section 5.4.1 finds the largest effect among children born after the first verified arrival but before the affected harvest was due, the Kenyan counterpart of the anticipatory window Conte et al. (2023) use in Mali. Unlike in Mali, local maize prices do not explain it. Controlling for them leaves the estimate unchanged, and the price premium in treated markets predates the first verified arrival. The post-boundary price fall more plausibly reflects COVID-era price disruptions than a supply response to crop failure. I therefore cannot separate a price-mediated channel, anticipatory or realized, from the direct loss of a household's own harvest. The health estimates capture whatever combination of the two operated.

The results have implications for policy. Among the 7,603 children with valid anthropometry, a relief program that defines exposure by proximity might enroll 4,070 children where a damage-verified rule enrolls 1,760, and the 2,310 children it adds are from drier, poorer, and more remote households that hold less land and are disproportionately pastoral. The two rules therefore select different populations. A household that lost a harvest needs aid sized to replace that harvest. A poor, remote pastoral household with no crop to

lose needs income support rather than compensation for a crop loss. A targeting rule based only on whether a swarm was nearby cannot distinguish the two, and it would direct harvest-replacement aid to households that had no harvest to replace while missing the different needs of pastoral communities.

For research, the results suggest that exposure measures for natural disasters should reflect how the hazard causes damage. These crises damage crops, assets, or health through a specific physical or biological process, and a measure built only on where the hazard was recorded does not capture that process.

Two extensions would add to these results. In the short run, more frequent surveys during a disaster could record how households respond. Knowing whether they sell livestock, change their diets or migrate, or keep harvests they would otherwise have sold would inform policy. In the longer run, following these cohorts could show whether the deficits measured here persist into cognitive development, schooling, and earnings.

A Supplemental Tables and Figures

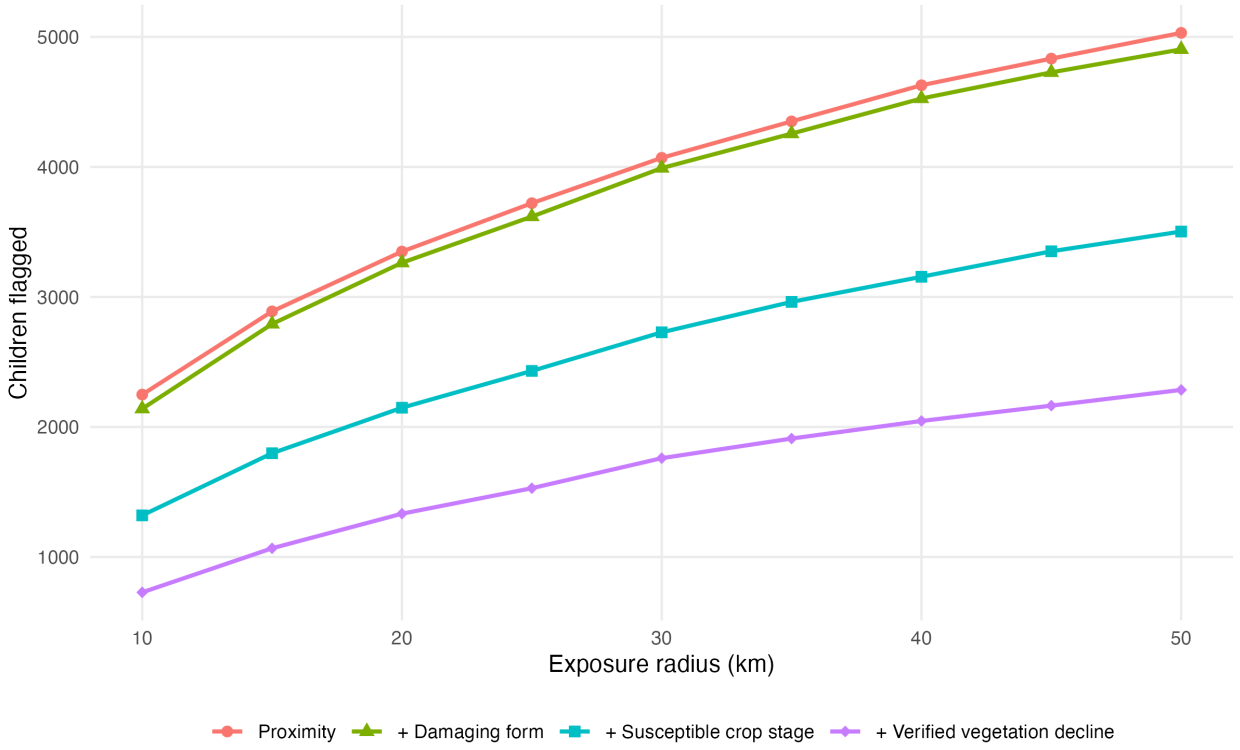


Figure 9: Children flagged as exposed, by exposure definition and radius. The four definitions nest, so every child flagged by a stricter measure is also flagged by every looser measure above it. Reproduces the counts of Table 9.

Table 21 compares exposed and unexposed children on the covariates available before treatment. The unconditional comparison looks like a typical cross-section of Kenyan households sorted by geography: exposed children have better-educated mothers, are more likely to own land, have higher wealth, and live closer to a road, and thirteen of the nineteen variables differ at the 5 percent level. Almost none of this survives conditioning on the same cluster and birth-cohort structure the main specification uses. Mother's education, wealth, land ownership, and household size all become indistinguishable from zero. Mother's height and household head age remain imbalanced after conditioning, both at the 10 percent level. Gestational NDVI remains significant after conditioning, in the opposite direction from its raw comparison, and gestational SPEI points the same way but is no longer significant ($p = 0.13$). Exposed children experience modestly less green, and somewhat drier, gestational windows than their unexposed neighbors within the same cluster and cohort. I address this in Section 5.3.

Table 6: Sensitivity of the exposure construction: weight-for-height z-score

	10 km		30 km		50 km	
Variant	Estimate	Treated	Estimate	Treated	Estimate	Treated
Main specification	−0.313***	698	−0.185***	1,703	−0.119*	2,210
	(0.084)		(0.069)		(0.062)	
NDVI decline threshold (main: $\theta = 0.25$)						
$\theta = 0$	−0.274***	815	−0.179***	1,957	−0.172***	2,512
	(0.075)		(0.068)		(0.063)	
$\theta = 0.1$	−0.286***	774	−0.180***	1,846	−0.161***	2,386
	(0.084)		(0.064)		(0.052)	
$\theta = 0.5$	−0.341***	563	−0.212***	1,426	−0.156***	1,891
	(0.080)		(0.071)		(0.059)	
$\theta = 0.75$	−0.297***	424	−0.180**	1,118	−0.118**	1,495
	(0.107)		(0.069)		(0.060)	
$\theta = 1$	−0.332***	297	−0.177***	805	−0.125**	1,128
	(0.112)		(0.056)		(0.054)	
$\theta = 1.5$	−0.290*	138	−0.213***	357	−0.082	530
	(0.154)		(0.067)		(0.075)	
NDVI composites before / after the event (main: 2 / 2)						
1 before / 1 after	−0.257***	657	−0.181**	1,759	−0.104	2,350
	(0.077)		(0.080)		(0.065)	
3 before / 3 after	−0.280***	724	−0.153**	1,783	−0.125**	2,294
	(0.060)		(0.060)		(0.049)	
2 before / 1 after	−0.332***	574	−0.211***	1,569	−0.169***	2,102
	(0.074)		(0.068)		(0.058)	
2 before / 3 after	−0.255***	765	−0.157***	1,830	−0.108**	2,355
	(0.073)		(0.058)		(0.051)	
1 before / 2 after	−0.199**	756	−0.134**	1,856	−0.103**	2,429
	(0.077)		(0.066)		(0.051)	
Crop calendar (main: onset-shifted, planting and standing crop susceptible)						
Standing crop only	−0.390***	415	−0.215**	1,055	−0.188***	1,340
	(0.126)		(0.089)		(0.069)	
Unshifted FEWS NET calendar	−0.284***	789	−0.107	1,839	−0.060	2,394
	(0.075)		(0.075)		(0.073)	
Stage windows widened 15 days	−0.273***	853	−0.095	2,000	−0.086	2,624
	(0.072)		(0.071)		(0.064)	
Stage windows narrowed 15 days	−0.285*	354	−0.168	911	−0.125	1,293
	(0.147)		(0.112)		(0.086)	
Gestation length (main: 266 days)						
280 days	−0.289***	739	−0.159***	1,795	−0.091	2,318
	(0.078)		(0.058)		(0.059)	
Sampling weights (main: DHS individual weight)						
Winsorized at 1st / 99th percentiles	−0.295***	698	−0.164***	1,703	−0.101*	2,210
	(0.076)		(0.059)		(0.055)	
Unweighted	−0.248***	698	−0.103**	1,703	−0.077	2,210
	(0.065)		(0.050)		(0.049)	

Each cell is a separate regression of the weight-for-height z-score on a rebuilt verified-exposure indicator, with the control set and fixed effects of column (3) of Table 2. Each variant changes one condition and holds the others at their main values: $\theta = 0.25$, two NDVI composites before and after the event, the onset-shifted calendar with planting and the standing crop susceptible, a 266-day gestation, and the DHS individual sampling weight. The standing-crop variant counts only the harvest (standing crop) stage as susceptible. The unshifted variant uses the transcribed FEWS NET stage dates without the rainfall-onset shift. The widened and narrowed variants move both ends of every planting and standing-crop interval out or in by 15 days. The 280-day variant extends the gestation window 14 days earlier. The winsorized variant caps the sampling weight at its 1st and 99th percentiles, and the unweighted variant drops it; both keep the main treatment indicator. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. The estimation sample is 7,416 children in every cell. Treated counts are children in the estimation sample.

Table 7: Sensitivity of the exposure construction: weight-for-age z-score

Variant	10 km	30 km	50 km
Main specification	−0.337***	−0.167**	−0.114*
	(0.103)	(0.069)	(0.061)
NDVI decline threshold (main: $\theta = 0.25$)			
$\theta = 0$	−0.281***	−0.154**	−0.141**
	(0.097)	(0.060)	(0.055)
$\theta = 0.1$	−0.293***	−0.145**	−0.130***
	(0.106)	(0.059)	(0.049)
$\theta = 0.5$	−0.361***	−0.184**	−0.148**
	(0.107)	(0.073)	(0.059)
$\theta = 0.75$	−0.317**	−0.155**	−0.105
	(0.142)	(0.072)	(0.066)
$\theta = 1$	−0.340**	−0.132**	−0.095
	(0.145)	(0.065)	(0.061)
$\theta = 1.5$	−0.324**	−0.163**	−0.025
	(0.132)	(0.080)	(0.065)
NDVI composites before / after the event (main: 2 / 2)			
1 before / 1 after	−0.267***	−0.173**	−0.117
	(0.097)	(0.081)	(0.075)
3 before / 3 after	−0.274***	−0.133**	−0.112**
	(0.081)	(0.058)	(0.045)
2 before / 1 after	−0.352***	−0.207***	−0.171**
	(0.089)	(0.075)	(0.066)
2 before / 3 after	−0.273***	−0.131**	−0.087*
	(0.099)	(0.059)	(0.051)
1 before / 2 after	−0.204**	−0.130*	−0.112*
	(0.098)	(0.066)	(0.060)
Crop calendar (main: onset-shifted, planting and standing crop susceptible)			
Standing crop only	−0.429***	−0.178*	−0.162**
	(0.151)	(0.093)	(0.072)
Unshifted FEWS NET calendar	−0.291***	−0.107	−0.071
	(0.099)	(0.077)	(0.071)
Stage windows widened 15 days	−0.271***	−0.112	−0.110*
	(0.097)	(0.072)	(0.064)
Stage windows narrowed 15 days	−0.317*	−0.158	−0.145*
	(0.175)	(0.108)	(0.081)
Gestation length (main: 266 days)			
280 days	−0.304***	−0.147***	−0.094*
	(0.079)	(0.053)	(0.053)
Sampling weights (main: DHS individual weight)			
Winsorized at 1st / 99th percentiles	−0.309***	−0.145**	−0.093*
	(0.087)	(0.058)	(0.052)
Unweighted	−0.222***	−0.063	−0.041
	(0.060)	(0.046)	(0.041)

Each cell is a separate regression of the weight-for-age z-score on a rebuilt verified-exposure indicator, with the control set and fixed effects of column (3) of Table 2. Each variant changes one condition and holds the others at their main values: $\theta = 0.25$, two NDVI composites before and after the event, the onset-shifted calendar with planting and the standing crop susceptible, a 266-day gestation, and the DHS individual sampling weight. The standing-crop variant counts only the harvest (standing crop) stage as susceptible. The unshifted variant uses the transcribed FEWS NET stage dates without the rainfall-onset shift. The widened and narrowed variants move both ends of every planting and standing-crop interval out or in by 15 days. The 280-day variant extends the gestation window 14 days earlier. The winsorized variant caps the sampling weight at its 1st and 99th percentiles, and the unweighted variant drops it; both keep the main treatment indicator. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. The estimation sample is 7,421 children in every cell. Treated counts differ by at most two children from those in Table 6.

Table 8: Sensitivity of the exposure construction: height-for-age z-score

Variant	10 km	30 km	50 km
Main specification	−0.193*	−0.059	−0.046
	(0.099)	(0.060)	(0.056)
NDVI decline threshold (main: $\theta = 0.25$)			
$\theta = 0$	−0.154	−0.048	−0.028
	(0.103)	(0.052)	(0.042)
$\theta = 0.1$	−0.152	−0.025	−0.022
	(0.104)	(0.051)	(0.044)
$\theta = 0.5$	−0.201	−0.053	−0.056
	(0.122)	(0.061)	(0.055)
$\theta = 0.75$	−0.183	−0.038	−0.030
	(0.144)	(0.069)	(0.071)
$\theta = 1$	−0.177	−0.007	−0.008
	(0.144)	(0.070)	(0.068)
$\theta = 1.5$	−0.236	−0.009	0.058
	(0.188)	(0.137)	(0.117)
NDVI composites before / after the event (main: 2 / 2)			
1 before / 1 after	−0.149	−0.057	−0.064
	(0.107)	(0.059)	(0.069)
3 before / 3 after	−0.129	−0.032	−0.022
	(0.102)	(0.055)	(0.046)
2 before / 1 after	−0.193*	−0.091	−0.084
	(0.100)	(0.065)	(0.063)
2 before / 3 after	−0.166	−0.024	−0.002
	(0.114)	(0.055)	(0.049)
1 before / 2 after	−0.106	−0.044	−0.053
	(0.106)	(0.052)	(0.065)
Crop calendar (main: onset-shifted, planting and standing crop susceptible)			
Standing crop only	−0.254*	−0.040	−0.036
	(0.129)	(0.083)	(0.069)
Unshifted FEWS NET calendar	−0.138	−0.051	−0.049
	(0.098)	(0.072)	(0.063)
Stage windows widened 15 days	−0.120	−0.072	−0.082
	(0.096)	(0.066)	(0.063)
Stage windows narrowed 15 days	−0.170	−0.044	−0.074
	(0.144)	(0.079)	(0.060)
Gestation length (main: 266 days)			
280 days	−0.172**	−0.061	−0.050
	(0.066)	(0.051)	(0.045)
Sampling weights (main: DHS individual weight)			
Winsorized at 1st / 99th percentiles	−0.165*	−0.046	−0.032
	(0.085)	(0.056)	(0.051)
Unweighted	−0.068	0.016	0.021
	(0.062)	(0.052)	(0.050)

Each cell is a separate regression of the height-for-age z-score on a rebuilt verified-exposure indicator, with the control set and fixed effects of column (3) of Table 2. Each variant changes one condition and holds the others at their main values: $\theta = 0.25$, two NDVI composites before and after the event, the onset-shifted calendar with planting and the standing crop susceptible, a 266-day gestation, and the DHS individual sampling weight. The standing-crop variant counts only the harvest (standing crop) stage as susceptible. The unshifted variant uses the transcribed FEWS NET stage dates without the rainfall-onset shift. The widened and narrowed variants move both ends of every planting and standing-crop interval out or in by 15 days. The 280-day variant extends the gestation window 14 days earlier. The winsorized variant caps the sampling weight at its 1st and 99th percentiles, and the unweighted variant drops it; both keep the main treatment indicator. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. The estimation sample is 7,424 children in every cell. Treated counts differ by at most two children from those in Table 6.

Table 9: Children flagged as exposed, by measure and radius

Radius (km)	Proximity	+ form	+ stage	+ verified	Disputed	Jaccard
10	2,249	2,140	1,320	728	1,521	0.32
15	2,889	2,792	1,798	1,067	1,822	0.37
20	3,349	3,263	2,148	1,333	2,016	0.40
25	3,721	3,618	2,431	1,529	2,192	0.41
30	4,070	3,991	2,728	1,760	2,310	0.43
35	4,350	4,256	2,962	1,911	2,439	0.44
40	4,628	4,526	3,155	2,046	2,582	0.44
45	4,833	4,727	3,351	2,164	2,669	0.45
50	5,031	4,904	3,503	2,285	2,746	0.45

Sample is the 7,603 children with valid anthropometry. Measures nest, so no child is flagged by a stricter measure and not by a looser one. Disputed denotes children flagged by proximity and not by the verified measure. Jaccard is the overlap between the proximity and verified lists.

Table 10: Dose response among verified-exposed children: number of damage events and height-for-age

Dependent variable: Height-for-age z-score	Linear	Log	Capped at 20
Verified damage events	-0.011^{*}		
	(0.006)		
Log(1 + verified damage events)		-0.181^{**}	
		(0.088)	
Verified damage events, capped at 20			-0.025^{**}
			(0.011)
Observations	1,460	1,460	1,460

Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Sample restricted to children verified-exposed at the main 30 km radius; of the 1,758 verified-exposed children with complete data, those who are the only such child in their survey cluster or birth cohort drop out of the fixed-effects estimation, which leaves 1,460. All columns include the control set and the survey-cluster and birth-cohort fixed effects of column (3) of Table 14. I control for the number of RAMSES ground visits during gestation, counting every recorded visit within 30 km, of any outcome, during the child's gestation window, to separate the effect of confirmed damage from the effect of monitoring intensity in that area during that period.

Table 11: Dose response among verified-exposed children: number of damage events and weight-for-age

Dependent variable: Weight-for-age z-score	Linear	Log	Capped at 20
Verified damage events	-0.016^{***}		
	(0.006)		
Log(1 + verified damage events)		-0.279^{***}	
		(0.056)	
Verified damage events, capped at 20			-0.037^{***}
			(0.006)
Observations	1,459	1,459	1,459

Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Sample restricted to children verified-exposed at the main 30 km radius; of the 1,757 verified-exposed children with complete data, those who are the only such child in their survey cluster or birth cohort drop out of the fixed-effects estimation, which leaves 1,459. All columns include the control set and the survey-cluster and birth-cohort fixed effects of column (3) of Table 15. I control for the number of RAMSES ground visits during gestation, counting every recorded visit within 30 km, of any outcome, during the child's gestation window, to separate the effect of confirmed damage from the effect of monitoring intensity in that area during that period.

Table 12: Alternative Anthropometric Outcomes, 30 km radius

	Height-for-age z-score	Weight-for-age z-score	Weight-for-height z-score	Stunted	Underweight	Wasted
In utero exposure	−0.059	−0.167**	−0.185***	0.006	−0.002	0.019**
	(0.060)	(0.069)	(0.069)	(0.016)	(0.012)	(0.008)
Baseline prevalence				0.185	0.122	0.072
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Cluster and birth-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,424	7,421	7,416	7,424	7,421	7,416
R^2	0.390	0.377	0.352	0.319	0.275	0.261

Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Each column is a separate regression sharing the control set and fixed effects of column (3) of Table 2. Baseline prevalence is the share of unexposed children below the relevant threshold.

Table 13: In utero exposure by age at measurement, 30 km radius

Age at measurement	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score	Treated
0 to 11 months	−0.410**	−0.133	−0.400**	551
	(0.181)	(0.206)	(0.203)	
12 to 23 months	−0.200**	−0.057	−0.170	1,035
	(0.091)	(0.132)	(0.108)	
24 months or older	0.122	−0.609**	−0.225	174
	(0.461)	(0.292)	(0.346)	

Each cell is a separate regression on children in the row's age group, sharing the control set and fixed effects of column (3) of Table 2. Observations are 2,371, 3,007, and 711 for weight-for-height, 2,376, 3,012, and 712 for height-for-age, and 2,375, 3,009, and 712 for weight-for-age, by age group. Treated counts the children with valid anthropometry in each age group flagged by the verified measure at 30 km. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 14: In utero locust exposure and child stunting, 30 km radius

	Height-for-age z-score				Stunted	
	(1)	(2)	(3)	(4)	(5)	(6)
In utero exposure	−0.022	−0.035	−0.059	−0.070	0.006	0.020
	(0.048)	(0.051)	(0.060)	(0.053)	(0.016)	(0.016)
Child controls	No	Yes	Yes	Yes	Yes	Yes
Family controls	No	No	Yes	Yes	Yes	Yes
County time trend	No	No	No	Yes	No	Yes
Observations	7,451	7,451	7,424	7,424	7,424	7,424
R^2	0.335	0.355	0.390	0.396	0.319	0.327

None of the six columns is significant at conventional levels. All regressions include birth cohort and enumeration area fixed effects. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All regressions weight by the DHS individual sampling weight. Child controls are sex, birth order, first-born status, twin status, and the preceding birth interval. Family controls are mother's education, mother's height, household head age and sex, wealth-quintile indicators, and six-month gestational SPEI. County time trend denotes county-specific linear trends in calendar month of birth. The outcome in columns (5) and (6) is an indicator for a height-for-age z-score below −2, the clinical threshold for stunting.

Table 15: In utero locust exposure and child underweight status, 30 km radius

	Weight-for-age z-score				Underweight	
	(1)	(2)	(3)	(4)	(5)	(6)
In utero exposure	−0.156**	−0.166***	−0.167**	−0.190***	−0.002	0.015
	(0.062)	(0.063)	(0.069)	(0.070)	(0.012)	(0.013)
Child controls	No	Yes	Yes	Yes	Yes	Yes
Family controls	No	No	Yes	Yes	Yes	Yes
County time trend	No	No	No	Yes	No	Yes
Observations	7,448	7,448	7,421	7,421	7,421	7,421
R^2	0.343	0.361	0.377	0.385	0.275	0.291

The point estimate is negative and significant in columns (1)-(4) but the binary (underweight) columns (5)-(6) are not significant, consistent with dilution similar to Section 5's weight-for-age discussion at the main radius. All regressions include birth cohort and enumeration area fixed effects. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All regressions weight by the DHS individual sampling weight. Child controls are sex, birth order, first-born status, twin status, and the preceding birth interval. Family controls are mother's education, mother's height, household head age and sex, wealth-quintile indicators, and six-month gestational SPEI. County time trend denotes county-specific linear trends in calendar month of birth. The outcome in columns (5) and (6) is an indicator for a weight-for-age z-score below −2, the clinical threshold for underweight.

Table 16: Treatment by Radius and Health Outcome

Radius	Weight-for-height z-score	Weight-for-age z-score	Height-for-age z-score	Wasted	Treated
10 km	−0.313***	−0.337***	−0.193*	0.023**	698
	(0.084)	(0.103)	(0.099)	(0.012)	
15 km	−0.284**	−0.251***	−0.094*	0.028***	1,027
	(0.114)	(0.094)	(0.050)	(0.009)	
20 km	−0.219***	−0.215***	−0.116**	0.021**	1,288
	(0.079)	(0.079)	(0.058)	(0.009)	
25 km	−0.186**	−0.194***	−0.111**	0.021**	1,479
	(0.074)	(0.068)	(0.050)	(0.009)	
30 km	−0.185***	−0.167**	−0.059	0.019**	1,703
	(0.069)	(0.069)	(0.060)	(0.008)	
35 km	−0.137*	−0.125*	−0.044	0.016*	1,850
	(0.073)	(0.070)	(0.057)	(0.009)	
40 km	−0.135*	−0.134**	−0.057	0.015*	1,978
	(0.070)	(0.066)	(0.055)	(0.009)	
45 km	−0.141**	−0.140**	−0.060	0.016*	2,091
	(0.069)	(0.066)	(0.056)	(0.009)	
50 km	−0.119*	−0.114*	−0.046	0.013	2,210
	(0.062)	(0.061)	(0.056)	(0.009)	

Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Each cell is a separate regression sharing the control set and fixed effects of column (3) of Table 2. Sample sizes are 7,416 for weight-for-height and wasting, 7,421 for weight-for-age, and 7,424 for height-for-age. Treated counts are treated children in the weight-for-height estimation sample.

Table 17: Falsification tests: purged placebo estimates on weight-for-height, by radius and comparison group

Radius	Solitary/transiens, any stage	Solitary/transiens, stage and NDVI	Out-of-season damaging form
Panel A: Purged			
10 km	0.065 (0.103)	—	0.074 (0.073)
15 km	0.086 (0.091)	—	0.022 (0.060)
20 km	0.123 (0.078)	0.284 (0.256)	0.008 (0.075)
25 km	0.139* (0.077)	0.641** (0.267)	0.085 (0.058)
30 km	0.036 (0.103)	0.168 (0.324)	0.085 (0.062)
35 km	0.027 (0.088)	0.005 (0.226)	0.019 (0.064)
40 km	0.088 (0.064)	0.284 (0.181)	−0.045 (0.059)
45 km	0.078 (0.062)	0.201 (0.167)	−0.020 (0.057)
50 km	0.023 (0.072)	0.032 (0.125)	−0.051 (0.062)
Panel B: Purged, verified exposure controlled			
10 km	0.015 (0.101)	—	0.031 (0.074)
15 km	0.029 (0.095)	—	−0.038 (0.064)
20 km	0.080 (0.079)	0.237 (0.260)	−0.049 (0.078)
25 km	0.099 (0.078)	0.591** (0.276)	0.028 (0.063)
30 km	−0.011 (0.107)	0.112 (0.331)	0.019 (0.069)
35 km	−0.012 (0.095)	−0.040 (0.238)	−0.038 (0.073)
40 km	0.049 (0.070)	0.234 (0.181)	−0.115* (0.066)
45 km	0.034 (0.067)	0.140 (0.171)	−0.099* (0.058)
50 km	−0.026 (0.085)	−0.026 (0.137)	−0.128** (0.064)
Panel C: Purged, verified-exposed children dropped			
10 km	−0.021 (0.093)	—	0.053 (0.073)
15 km	0.033 (0.085)	—	−0.045 (0.066)
20 km	0.069 (0.074)	0.216 (0.297)	−0.064 (0.079)
25 km	0.036 (0.075)	0.463 (0.315)	0.027 (0.065)
30 km	−0.074 (0.119)	0.075 (0.406)	0.064 (0.076)
35 km	−0.072 (0.105)	−0.062 (0.295)	0.031 (0.091)
40 km	0.000 (0.077)	0.241 (0.245)	−0.049 (0.083)
45 km	−0.044 (0.073)	−0.001 (0.222)	−0.003 (0.078)
50 km	−0.161* (0.089)	−0.229* (0.117)	−0.036 (0.088)

Outcome is weight-for-height (WHZ). Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All three columns are purged of overlap with verified (D4) exposure at the same radius, as described in Section 5.3, and share the control set and fixed effects of column (3) of Table 2. Solitary/transiens, any stage is solitary and transiens locust records purged of D4 overlap; Solitarious/transiens, stage and NDVI additionally requires the crop-stage and vegetation-decline conditions of equation (9) and is not estimated at 10 or 15 km, where purging leaves fewer than 20 children (see Figure 10); Out-of-season damaging form is damaging-form records purged of D4 overlap that arrive outside the susceptible crop stage. In Panel A, verified-exposed children sit in the comparison group. Panel B adds the verified indicator at the same radius as a control. Panel C drops verified-exposed children from the sample. Each panel has 27 nominal cells, 25 estimated after the 20-observation floor.

Table 18: Vegetation decline without a locust record: pseudo-event placebo

	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score	Died	Share treated
Panel A: 2014 round					
Any decline	0.205	−0.270*	0.005	−0.034	0.81
	(0.125)	(0.158)	(0.122)	(0.024)	
Matched	0.005	0.032	0.017	0.003	0.23
	(0.041)	(0.062)	(0.041)	(0.009)	
Count	−0.002	0.025	0.011	−0.004*	—
	(0.016)	(0.024)	(0.018)	(0.002)	
Panel B: 2022 round					
Any decline	0.260*	−0.038	0.213	0.019	0.78
	(0.144)	(0.185)	(0.194)	(0.021)	
Matched	−0.161**	−0.003	−0.097*	−0.002	0.23
	(0.069)	(0.049)	(0.056)	(0.007)	
Count	−0.023	−0.038***	−0.036***	0.002	—
	(0.015)	(0.013)	(0.013)	(0.002)	

Every 16-day NDVI composite in a child's own cluster is treated as a candidate event and must satisfy the decline condition of equation (8) and the crop-stage condition of equation (5); no locust record is required. *Any decline* is an indicator for at least one qualifying decline during gestation. *Matched* is an indicator for the children whose largest decline during gestation is among the most severe in their round, with the cutoff (−1.59 standard deviations in 2014, −1.76 in 2022) set so that the treated share equals the verified measure's share at 30 km. *Count* is the number of qualifying declines during gestation. Each cell is a separate regression sharing the control set and fixed effects of column (3) of Table 2. Observations are 7,982 for the anthropometric outcomes and 8,698 for death in 2014, and 7,416, 7,424, 7,421, and 8,057 in 2022. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 19: Verified exposure and vegetation decline with and without a nearby locust record, weight-for-height, 2022

	(1)	(2)	(3)	(4)
Verified exposure	−0.184***	−0.188***		−0.167**
	(0.067)	(0.069)		(0.067)
Matched decline	−0.159**			
	(0.066)			
Matched decline, locust record within 50 km			−0.217***	−0.195***
			(0.077)	(0.074)
Matched decline, no locust record within 50 km		−0.022	−0.034	−0.079
		(0.065)	(0.073)	(0.071)
Observations	7,416	7,416	7,416	7,416

Outcome is the weight-for-height z-score. Verified exposure is the main 30 km measure. Matched decline is defined as in Table 18. It is split by whether any locust record of any form fell within 50 km of the child's cluster during gestation. All columns share the control set and fixed effects of column (3) of Table 2. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

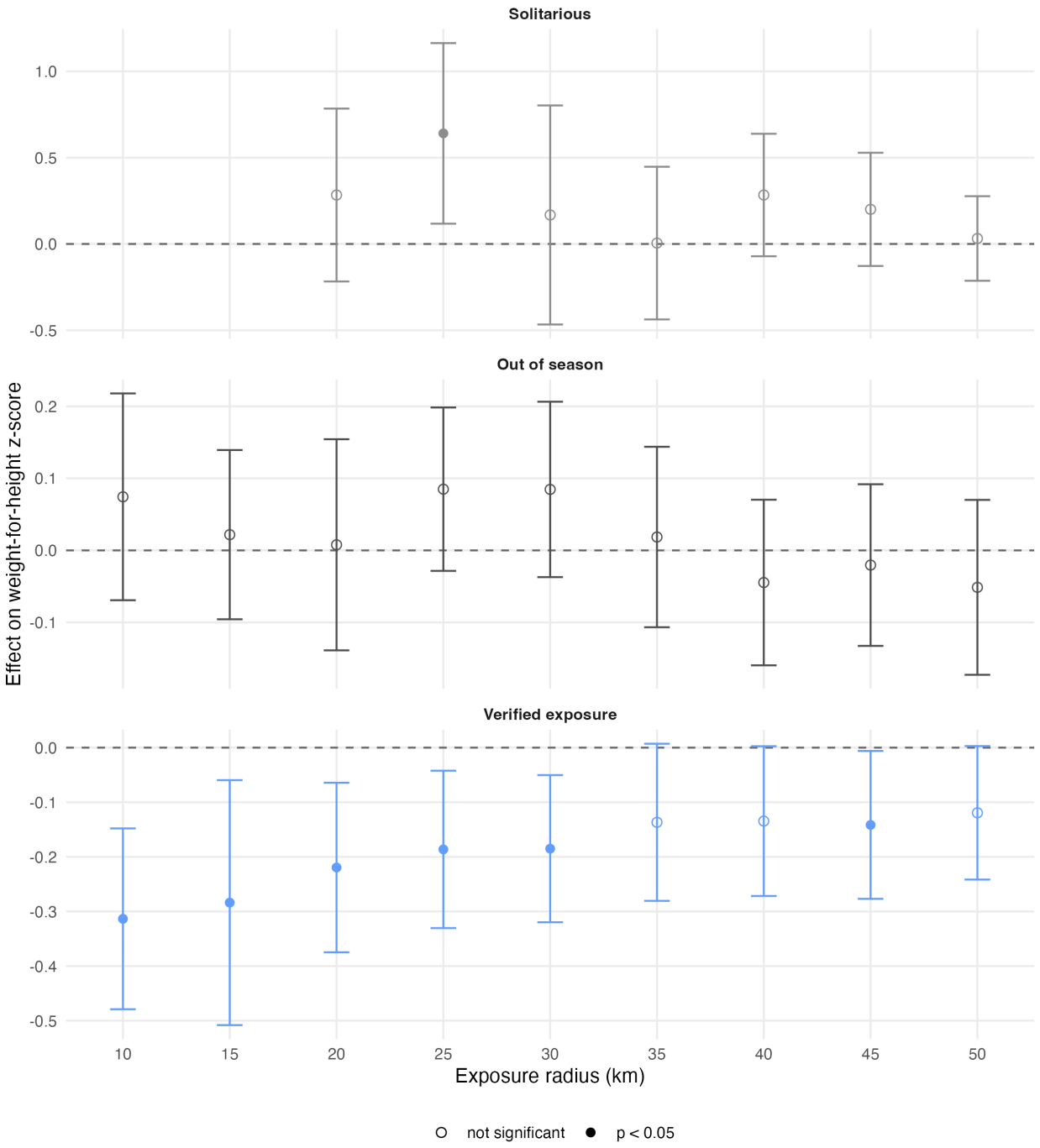


Figure 10: Falsification tests and verified exposure on weight-for-height, by radius. Solitarious and out-of-season estimates are purged of overlap with verified exposure at the same radius, as described in Section 5.3. Filled points are significant at the 5 percent level; hollow points are not. Y-axes are independent across panels and are not comparable to one another in scale. No solitarious estimate, which applies the stage and vegetation conditions, is reported at 10 or 15 km: purging leaves 11 and 19 children respectively which are both below a 20-observation floor I apply for this test.

Table 20: Count of qualifying vegetation declines, by whether a locust record fell nearby, 2022

	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score
Panel A: Count split by nearby locust record			
Declines, locust record within 50 km	−0.027*	−0.039***	−0.040***
	(0.015)	(0.012)	(0.013)
Declines, no locust record within 50 km	−0.016	−0.035*	−0.029**
	(0.016)	(0.018)	(0.014)
Panel B: Adding verified exposure			
Verified exposure	−0.177**	−0.037	−0.146**
	(0.075)	(0.059)	(0.072)
Declines, locust record within 50 km	−0.018	−0.037***	−0.032**
	(0.014)	(0.012)	(0.012)
Declines, no locust record within 50 km	−0.018	−0.036**	−0.031**
	(0.017)	(0.018)	(0.014)
Observations	7,416	7,424	7,421

The count of qualifying declines of Table 18 is split by whether any locust record of any form fell within 50 km of the child's cluster during gestation. Each panel and column is a single regression sharing the control set and fixed effects of column (3) of Table 2. Verified exposure is the main 30 km measure. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 21: Covariate balance, 30 km radius

	Control	Treatment	Raw difference	FE-adjusted	(SE)
Child					
Female	0.490	0.476	−0.014	−0.026	(0.024)
Birth order	3.38	2.75	−0.629***	−0.054	(0.085)
First born	0.242	0.309	0.066***	0.022	(0.028)
Twin	0.031	0.021	−0.010**		—
Birth interval, prior (months)	46.0	52.8	6.74***	1.53	(1.85)
Birth interval, next (months)	18.5	16.8	−1.70***		—
Mother and household					
Mother's education (years)	7.13	9.42	2.30***	−0.045	(0.161)
Mother's height (cm)	160.8	160.6	−0.184	0.342*	(0.205)
Household head age	38.9	38.3	−0.576*	−1.04*	(0.608)
Female household head	0.291	0.268	−0.023*	−0.020	(0.022)
Household size	6.06	5.38	−0.683***	−0.020	(0.141)
Wealth quintile	2.48	3.00	0.519***	0.021	(0.032)
Assets and access					
Owns land	0.462	0.545	0.082***	−0.014	(0.017)
Owns livestock	0.687	0.662	−0.025*	0.018	(0.014)
Land owned (hectares)	1.07	0.83	−0.238	−0.380	(0.408)
Rural	0.675	0.640	−0.034***		—
Distance to major road (km)	8.34	3.75	−4.59***		—
Gestational climate					
SPEI (6 month)	0.997	1.265	0.268***	−0.177	(0.117)
NDVI z-score	0.678	0.747	0.070***	−0.164***	(0.046)
Observations	5,843	1,760			

Raw difference is the unconditional mean difference, exposed minus unexposed, with significance from a two-sample t-test. FE-adjusted is the coefficient on the exposure indicator from regressing each covariate on exposure with the same cluster and birth-cohort fixed effects, sampling weight, and grid-clustered standard error as column (3) of Table 2. ***, **, * denote significance at 1, 5 and 10 percent. Rural and distance to a major road are fixed within a DHS cluster and have no within-cluster variation once the cluster fixed effect is included, so no FE-adjusted coefficient is reported for either. Twin status and the birth interval to the next child likewise return no FE-adjusted coefficient in the same cluster-by-cohort structure, so none is reported for either. Counts in the observation block are the 7,603 children with valid anthropometry; the estimation sample for each covariate differs where a covariate has missing values.

Table 22: Weight-for-height estimates with and without the drought control

Radius	With SPEI-6	Without SPEI-6	Treated
10 km	−0.313***	−0.320***	698
	(0.084)	(0.082)	
15 km	−0.284**	−0.289**	1,027
	(0.114)	(0.111)	
20 km	−0.219***	−0.225***	1,288
	(0.079)	(0.077)	
25 km	−0.186**	−0.192***	1,479
	(0.074)	(0.071)	
30 km	−0.185***	−0.191***	1,703
	(0.069)	(0.068)	
35 km	−0.137*	−0.144**	1,850
	(0.073)	(0.072)	
40 km	−0.135*	−0.142**	1,978
	(0.070)	(0.069)	
45 km	−0.141**	−0.149**	2,091
	(0.069)	(0.068)	
50 km	−0.119*	−0.127**	2,210
	(0.062)	(0.062)	

Outcome is the weight-for-height z-score. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. *With SPEI-6* repeats the specification of column (3) of Table 2. *Without SPEI-6* is identical except for dropping the six-month gestational SPEI control, the only variable in the specification that measures drought directly. Observations are 7,416 in every cell. Treated counts are the treated children in that estimation sample, which excludes children with missing data and children who are the only child in their cluster.

Table 23: Exposure and Mortality

Radius	Estimate	Treated	p
10 km	0.003 (0.009)	792	0.727
15 km	0.001 (0.008)	1,160	0.870
20 km	0.005 (0.006)	1,449	0.371
25 km	0.002 (0.006)	1,656	0.687
30 km	0.002 (0.006)	1,904	0.701
35 km	−0.001 (0.006)	2,066	0.853
40 km	0.005 (0.006)	2,209	0.467
45 km	0.004 (0.006)	2,338	0.529
50 km	0.003 (0.007)	2,467	0.614

Outcome is an indicator for death before interview. Standard errors clustered on a half-degree spatial grid in parentheses. Sample is 8,057 of the 8,213 births in the 2022 round, including children who died or were not measured (Section 5.3). Treated counts are among all 8,213 births. Baseline mortality is 3.2 percent. Each row is a separate regression sharing the control set and fixed effects of column (3) of Table 2.

Table 24: Shifted-calendar 2014 pre-invasion falsification: height-for-age z-score

Radius	Proximity	+ Damaging form	+ Susceptible crop stage	+ Verified vegetation decline
10 km	−0.092	−0.088	−0.095	0.031
	(0.085)	(0.086)	(0.089)	(0.114)
	[2,239]	[2,183]	[1,355]	[447]
15 km	−0.022	−0.012	−0.064	−0.022
	(0.111)	(0.113)	(0.109)	(0.116)
	[2,874]	[2,798]	[1,753]	[594]
20 km	0.001	0.009	−0.065	−0.052
	(0.107)	(0.108)	(0.108)	(0.134)
	[3,292]	[3,212]	[2,030]	[704]
25 km	0.071	0.066	−0.036	0.054
	(0.084)	(0.080)	(0.095)	(0.096)
	[3,746]	[3,647]	[2,359]	[864]
30 km	0.029	0.024	−0.041	0.000
	(0.087)	(0.086)	(0.101)	(0.076)
	[4,080]	[3,978]	[2,605]	[1,005]
35 km	0.059	0.054	−0.015	0.086
	(0.073)	(0.069)	(0.082)	(0.066)
	[4,436]	[4,334]	[2,896]	[1,164]
40 km	0.047	0.053	−0.039	0.042
	(0.070)	(0.066)	(0.085)	(0.067)
	[4,768]	[4,659]	[3,127]	[1,307]
45 km	0.060	0.074	−0.026	0.108
	(0.066)	(0.060)	(0.071)	(0.076)
	[4,965]	[4,839]	[3,285]	[1,408]
50 km	0.070	0.077	−0.026	0.123
	(0.065)	(0.058)	(0.064)	(0.082)
	[5,175]	[5,043]	[3,445]	[1,523]

Each cell is a separate regression of the height-for-age z-score on the exposure measure in the column, at the radius in the row, when the real invasion's calendar pattern is shifted back 93 months onto the 2014 DHS cohort's actual gestation windows (Section 5.3). Columns add the conditions of Section 4.1 one at a time, so the last column is the verified measure. Every regression shares the control set and fixed effects of column (3) of Table 2. Standard errors clustered on a half-degree spatial grid in parentheses; treated children in brackets. ***, **, * denote significance at 1, 5 and 10 percent. Across Tables 24 to 27, none of the 144 cells is significant at the 5 percent level and 5 (3.5 percent) are significant at the 10 percent level: the verified measure on weight-for-height at 10 km, the verified measure on mortality at 30, 35 and 50 km, and raw proximity on weight-for-age at 50 km.

Table 25: Shifted-calendar 2014 pre-invasion falsification: weight-for-age z-score

Radius	Proximity	+ Damaging form	+ Susceptible crop stage	+ Verified vegetation decline
10 km	−0.041	−0.051	−0.053	−0.073
	(0.049)	(0.049)	(0.050)	(0.078)
	[2,239]	[2,183]	[1,355]	[447]
15 km	−0.004	−0.017	−0.051	−0.057
	(0.063)	(0.062)	(0.061)	(0.081)
	[2,874]	[2,798]	[1,753]	[594]
20 km	0.009	−0.001	−0.040	−0.031
	(0.090)	(0.090)	(0.087)	(0.083)
	[3,292]	[3,212]	[2,030]	[704]
25 km	0.030	0.017	−0.025	0.030
	(0.064)	(0.064)	(0.062)	(0.065)
	[3,746]	[3,647]	[2,359]	[864]
30 km	0.007	−0.012	−0.035	0.001
	(0.069)	(0.069)	(0.068)	(0.055)
	[4,080]	[3,978]	[2,605]	[1,005]
35 km	0.037	0.014	−0.006	0.074
	(0.054)	(0.055)	(0.054)	(0.057)
	[4,436]	[4,334]	[2,896]	[1,164]
40 km	0.017	−0.003	−0.015	0.044
	(0.048)	(0.049)	(0.049)	(0.057)
	[4,768]	[4,659]	[3,127]	[1,307]
45 km	0.064	0.044	0.025	0.105
	(0.041)	(0.042)	(0.050)	(0.073)
	[4,965]	[4,839]	[3,285]	[1,408]
50 km	0.089*	0.056	0.029	0.124
	(0.046)	(0.042)	(0.049)	(0.076)
	[5,175]	[5,043]	[3,445]	[1,523]

Each cell is a separate regression of the weight-for-age z-score on the exposure measure in the column, at the radius in the row, when the real invasion's calendar pattern is shifted back 93 months onto the 2014 DHS cohort's actual gestation windows (Section 5.3). Columns add the conditions of Section 4.1 one at a time, so the last column is the verified measure. Every regression shares the control set and fixed effects of column (3) of Table 2. Standard errors clustered on a half-degree spatial grid in parentheses; treated children in brackets. ***, **, * denote significance at 1, 5 and 10 percent.

Table 26: Shifted-calendar 2014 pre-invasion falsification: weight-for-height z-score

Radius	Proximity	+ Damaging form	+ Susceptible crop stage	+ Verified vegetation decline
10 km	0.002	−0.016	−0.016	−0.162*
	(0.050)	(0.045)	(0.047)	(0.096)
	[2,239]	[2,183]	[1,355]	[447]
15 km	0.012	−0.013	−0.024	−0.088
	(0.057)	(0.047)	(0.053)	(0.086)
	[2,874]	[2,798]	[1,753]	[594]
20 km	0.010	−0.009	−0.010	−0.023
	(0.070)	(0.063)	(0.063)	(0.078)
	[3,292]	[3,212]	[2,030]	[704]
25 km	−0.009	−0.026	−0.011	−0.003
	(0.062)	(0.054)	(0.051)	(0.084)
	[3,746]	[3,647]	[2,359]	[864]
30 km	−0.011	−0.033	−0.017	−0.006
	(0.061)	(0.053)	(0.049)	(0.054)
	[4,080]	[3,978]	[2,605]	[1,005]
35 km	0.009	−0.019	0.000	0.036
	(0.055)	(0.049)	(0.050)	(0.055)
	[4,436]	[4,334]	[2,896]	[1,164]
40 km	−0.010	−0.040	0.007	0.028
	(0.056)	(0.050)	(0.052)	(0.061)
	[4,768]	[4,659]	[3,127]	[1,307]
45 km	0.045	0.007	0.050	0.062
	(0.057)	(0.053)	(0.068)	(0.073)
	[4,965]	[4,839]	[3,285]	[1,408]
50 km	0.077	0.028	0.063	0.085
	(0.060)	(0.048)	(0.062)	(0.071)
	[5,175]	[5,043]	[3,445]	[1,523]

Each cell is a separate regression of the weight-for-height z-score on the exposure measure in the column, at the radius in the row, when the real invasion's calendar pattern is shifted back 93 months onto the 2014 DHS cohort's actual gestation windows (Section 5.3). Columns add the conditions of Section 4.1 one at a time, so the last column is the verified measure. Every regression shares the control set and fixed effects of column (3) of Table 2. Standard errors clustered on a half-degree spatial grid in parentheses; treated children in brackets. ***, **, * denote significance at 1, 5 and 10 percent.

Table 27: Shifted-calendar 2014 pre-invasion falsification: mortality

Radius	Proximity	+ Damaging form	+ Susceptible crop stage	+ Verified vegetation decline
10 km	0.005	0.003	−0.001	−0.018
	(0.009)	(0.008)	(0.009)	(0.023)
	[2,416]	[2,354]	[1,461]	[474]
15 km	0.014	0.010	0.006	−0.013
	(0.009)	(0.009)	(0.010)	(0.020)
	[3,109]	[3,025]	[1,892]	[629]
20 km	0.011	0.008	0.003	−0.015
	(0.009)	(0.008)	(0.009)	(0.020)
	[3,562]	[3,474]	[2,189]	[748]
25 km	0.011	0.008	0.004	−0.012
	(0.009)	(0.008)	(0.009)	(0.014)
	[4,058]	[3,950]	[2,545]	[917]
30 km	0.012	0.009	0.005	−0.019*
	(0.008)	(0.008)	(0.009)	(0.010)
	[4,434]	[4,321]	[2,826]	[1,070]
35 km	0.005	0.002	−0.003	−0.024*
	(0.011)	(0.010)	(0.012)	(0.013)
	[4,810]	[4,695]	[3,134]	[1,237]
40 km	0.013	0.009	0.004	−0.010
	(0.009)	(0.008)	(0.010)	(0.011)
	[5,169]	[5,047]	[3,386]	[1,397]
45 km	0.009	0.002	−0.003	−0.017
	(0.012)	(0.011)	(0.013)	(0.014)
	[5,387]	[5,243]	[3,556]	[1,501]
50 km	0.005	0.004	−0.001	−0.021*
	(0.012)	(0.010)	(0.012)	(0.012)
	[5,608]	[5,458]	[3,724]	[1,625]

Each cell is a separate regression of an indicator for death before interview, over all 2014 births alive or dead, on the exposure measure in the column, at the radius in the row, when the real invasion's calendar pattern is shifted back 93 months onto the 2014 DHS cohort's actual gestation windows (Section 5.3). Columns add the conditions of Section 4.1 one at a time, so the last column is the verified measure. Every regression shares the control set and fixed effects of column (3) of Table 2. Standard errors clustered on a half-degree spatial grid in parentheses; treated children in brackets. ***, **, * denote significance at 1, 5 and 10 percent.

Table 28: In utero and postnatal exposure, weight-for-height, 30 km radius

	(1)	(2)	(3)	(4)
In utero exposure	−0.185***	−0.145*	−0.145*	
	(0.069)	(0.079)	(0.080)	
Postnatal exposure, first 12 months		0.106		0.138*
		(0.072)		(0.082)
Postnatal exposure, months 0 to 6			0.116*	
			(0.067)	
Postnatal exposure, months 6 to 12			0.084	
			(0.080)	
Sample	All	All	All	Not exposed in utero
Observations	7,416	7,416	7,416	5,513

Outcome is the weight-for-height z-score. Postnatal exposure is a verified damage event within 30 km of the child's cluster after birth, within the stated months and before the interview. All columns share the control set and fixed effects of column (3) of Table 2. Column (4) restricts to the 5,713 children in the estimation sample who were not exposed in utero; 200 of them are the only such child in their survey cluster or birth cohort and drop out of the cluster fixed-effects estimation, which leaves 5,513. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 29: Verified exposure and the treatment of missing mother's height

	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score	Treated
10 km				
Main specification	−0.313***	−0.193*	−0.337***	698
	(0.084)	(0.099)	(0.103)	
Adding a missing-height indicator	−0.314***	−0.192*	−0.337***	698
	(0.085)	(0.097)	(0.103)	
Dropping mother's height	−0.312***	−0.168*	−0.323***	698
	(0.083)	(0.089)	(0.095)	
Measured mothers only	−0.273**	−0.140	−0.268**	309
	(0.108)	(0.120)	(0.106)	
30 km				
Main specification	−0.185***	−0.059	−0.167**	1,703
	(0.069)	(0.060)	(0.069)	
Adding a missing-height indicator	−0.184***	−0.060	−0.167**	1,703
	(0.069)	(0.061)	(0.069)	
Dropping mother's height	−0.184***	−0.042	−0.157**	1,703
	(0.068)	(0.053)	(0.066)	
Measured mothers only	−0.174**	0.037	−0.112	763
	(0.079)	(0.089)	(0.080)	
50 km				
Main specification	−0.119*	−0.046	−0.114*	2,210
	(0.062)	(0.056)	(0.061)	
Adding a missing-height indicator	−0.119*	−0.047	−0.114*	2,210
	(0.063)	(0.056)	(0.062)	
Dropping mother's height	−0.118*	−0.025	−0.102*	2,210
	(0.062)	(0.051)	(0.059)	
Measured mothers only	−0.133	0.074	−0.086	986
	(0.082)	(0.082)	(0.082)	

Each cell is a separate regression with the control set and fixed effects of column (3) of Table 2. The main specification fills a missing mother's height with the median among measured women. The second row adds an indicator for a filled value, the third drops mother's height from the controls, and the fourth restricts the sample to children whose mother was measured (3,513 for weight-for-height, 3,516 for height-for-age and 3,514 for weight-for-age, against 7,416, 7,424 and 7,421). Treated counts are for the weight-for-height estimation sample. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 30: Off-season damaging-form exposure in cropping zones

Radius	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score	Treated
Panel A: Comparison group is every other child				
10 km	0.071	0.160	0.123	111
	(0.101)	(0.102)	(0.097)	
15 km	0.043	−0.008	0.019	115
	(0.144)	(0.116)	(0.142)	
20 km	−0.177	0.010	−0.116	108
	(0.165)	(0.090)	(0.122)	
25 km	0.063	−0.179	−0.044	129
	(0.106)	(0.126)	(0.089)	
30 km	0.240*	−0.142	0.093	139
	(0.131)	(0.107)	(0.112)	
35 km	0.063	−0.113	−0.020	116
	(0.147)	(0.132)	(0.133)	
40 km	−0.138	−0.204*	−0.213	124
	(0.152)	(0.120)	(0.135)	
45 km	−0.125	−0.182	−0.186	112
	(0.173)	(0.126)	(0.167)	
50 km	−0.240	−0.248**	−0.303*	121
	(0.169)	(0.114)	(0.153)	
Panel B: Stage and verified indicators controlled				
10 km	0.033	0.151	0.089	111
	(0.102)	(0.096)	(0.098)	
15 km	−0.006	−0.032	−0.030	115
	(0.144)	(0.114)	(0.144)	
20 km	−0.236	−0.014	−0.171	108
	(0.169)	(0.094)	(0.121)	
25 km	−0.041	−0.224	−0.148	129
	(0.116)	(0.141)	(0.102)	
30 km	0.205	−0.166	0.050	139
	(0.140)	(0.130)	(0.120)	
35 km	0.026	−0.132	−0.064	116
	(0.154)	(0.141)	(0.130)	
40 km	−0.201	−0.260*	−0.300**	124
	(0.155)	(0.133)	(0.134)	
45 km	−0.154	−0.210	−0.233	112
	(0.170)	(0.136)	(0.165)	
50 km	−0.274	−0.281**	−0.359**	121
	(0.170)	(0.128)	(0.150)	
Panel C: Stage- and verified-exposed children dropped				
10 km	0.168	0.142	0.180	111
	(0.102)	(0.149)	(0.122)	
15 km	−0.004	−0.039	0.000	115
	(0.161)	(0.127)	(0.164)	
20 km	−0.045	0.120	0.048	108
	(0.140)	(0.121)	(0.107)	
25 km	0.188	−0.310*	−0.048	129
	(0.171)	(0.183)	(0.099)	
30 km	0.514**	−0.269	0.241	139
	(0.239)	(0.203)	(0.160)	
35 km	0.329	−0.265	0.092	116
	(0.209)	(0.203)	(0.140)	
40 km	0.030	−0.282	−0.137	124
	(0.151)	(0.192)	(0.101)	
45 km	0.059	−0.221	−0.066	112
	(0.142)	(0.178)	(0.138)	
50 km	−0.048	−0.251	−0.148	121
	(0.206)	(0.154)	(0.173)	

Sample is children in cropping livelihood zones (4,552 for weight-for-height, 4,557 for height-for-age and 4,555 for weight-for-age in Panels A and B). The exposure indicator equals one for a child with a damaging-form record within the radius during gestation and no such record in a susceptible crop stage, so the child is flagged by the damaging-form measure but by neither the stage nor the verified measure. In Panel A the comparison group is every other child in the sample, including children treated under the stage or verified measures. Panel B adds the stage and verified indicators at the same radius as controls. Panel C drops children flagged by the stage measure, which includes every verified-exposed child, leaving 1,667 to 3,393 children depending on the radius and outcome. Each cell is a separate regression with the control set and fixed effects of column (3) of Table 2. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 31: Timing of gestation relative to the harvest boundary, 30 km radius

	Weight-for-height z-score		Height-for-age z-score		
	(1)	(2)	(3)	(4)	Treated
Anticipatory	−0.297***	−0.301***	−0.156	−0.153	613
	(0.110)	(0.111)	(0.097)	(0.099)	
Both	−0.150	−0.154	−0.028	−0.025	1,422
	(0.096)	(0.095)	(0.091)	(0.093)	
Crop failure	0.002	−0.006	−0.171	−0.164	1,174
	(0.107)	(0.109)	(0.129)	(0.136)	
Pastoral or fishing zone, proximity	−0.019	−0.021	−0.084	−0.083	1,023
	(0.098)	(0.098)	(0.101)	(0.101)	
Maize price control	No	Yes	No	Yes	
Observations	7,416	7,416	7,424	7,424	

Each column is a single regression of the column outcome on the four groups jointly, with the control set and fixed effects of column (3) of Table 2. Groups are defined from the first verified damage event within 30 km of the child's cluster and the harvest boundary of that event's crop, among children born on or after the event date. *Anticipatory*: born before the boundary. *Both*: the boundary falls during gestation. *Crop failure*: conceived after the boundary. *Pastoral or fishing zone, proximity*: children in pastoral and fishing livelihood zones with a locust record of any form within 30 km during gestation. Columns (2) and (4) add the gestational maize price index. Treated counts are for the weight-for-height estimation sample. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 32: Exposed children who are underweight versus exposed children who are not, 30 km radius

	Underweight	Not underweight	Difference	p
Mother's education (years)	6.88	9.72	−2.85	< 0.001
Wealth quintile	2.24	3.09	−0.85	< 0.001
Distance to major road (km)	5.15	3.58	1.57	0.001
Rural	0.757	0.626	0.130	< 0.001
Owns land	0.503	0.550	−0.048	0.218
Mother's height (cm)	159.54	160.75	−1.20	< 0.001
Gestational SPEI (6 month)	1.221	1.270	−0.049	0.522
Gestational NDVI z-score	0.877	0.731	0.145	0.015
Age at measurement (months)	15.82	15.66	0.16	0.737

Sample restricted to children flagged as exposed at 30 km. Underweight denotes a weight-for-age z-score below −2 ($N = 189$). Not underweight denotes the remainder of the exposed group ($N = 1{,}570$). p-values from two-sample t-tests.

Table 33: Exposed cases versus exposed non-cases, 15 km radius

	Case	Non-case	Difference	p
Panel A: Wasted versus not wasted				
Mother's education (years)	6.20	9.71	−3.51	< 0.001
Wealth quintile	2.29	3.20	−0.91	< 0.001
Distance to major road (km)	5.63	3.42	2.21	0.017
Rural	0.646	0.561	0.085	0.175
Owns land	0.431	0.523	−0.093	0.151
Mother's height (cm)	160.58	160.63	−0.05	0.926
Gestational SPEI (6 month)	1.086	1.216	−0.129	0.342
Gestational NDVI z-score	0.705	0.735	−0.030	0.754
Age at measurement (months)	14.68	15.70	−1.02	0.208
Panel B: Underweight versus not underweight				
Mother's education (years)	6.69	9.86	−3.17	< 0.001
Wealth quintile	2.31	3.26	−0.95	< 0.001
Distance to major road (km)	5.50	3.31	2.19	< 0.001
Rural	0.711	0.549	0.162	< 0.001
Owns land	0.496	0.520	−0.024	0.618
Mother's height (cm)	159.59	160.77	−1.18	0.004
Gestational SPEI (6 month)	1.189	1.210	−0.022	0.826
Gestational NDVI z-score	0.865	0.717	0.148	0.057
Age at measurement (months)	16.10	15.57	0.53	0.393

Sample restricted to children flagged as exposed at 15 km. Panel A compares children with a weight-for-height z-score below −2 ($N = 65$) to the remainder of the exposed group ($N = 1{,}001$). Panel B compares children with a weight-for-age z-score below −2 ($N = 121$) to the remainder ($N = 946$). p-values from two-sample t-tests.

Table 34: Heterogeneity: baseline characteristics entered jointly

	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score
Agricultural occupation	0.162	0.015	0.112
	(0.129)	(0.130)	(0.137)
Owns land	0.030	−0.031	−0.002
	(0.132)	(0.077)	(0.129)
Rural	0.206	0.073	0.199
	(0.165)	(0.090)	(0.144)
Distance to major road (km)	0.000	0.016	0.008
	(0.010)	(0.010)	(0.010)

Each column is a single regression of the column outcome on exposure at the main radius, all four baseline characteristics, their interactions with exposure, and the control set and fixed effects of column (3) of Table 2; each cell reports one characteristic's exposure interaction from that shared regression, so the four rows in a column are directly comparable. Compare to the one-at-a-time version in the top panel of Table 5: no characteristic is significant once the other three are controlled for directly. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 35: Riverine-versus-cropping exposure gap, by radius

Radius (km)	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score	Treated in riverine
10	−0.201	−0.211	−0.236	23
	(0.144)	(0.141)	(0.162)	
15	−0.222	−0.219	−0.264*	28
	(0.155)	(0.149)	(0.159)	
20	−0.268**	−0.196	−0.288*	28
	(0.129)	(0.151)	(0.150)	
25	−0.298**	−0.204	−0.309**	28
	(0.126)	(0.147)	(0.145)	
30	−0.246*	−0.390**	−0.369***	35
	(0.137)	(0.182)	(0.136)	
35	−0.291**	−0.406**	−0.410***	35
	(0.141)	(0.182)	(0.138)	
40	0.055	−0.058	−0.001	38
	(0.165)	(0.205)	(0.182)	
45	0.059	−0.056	0.001	38
	(0.163)	(0.204)	(0.181)	
50	0.115	−0.047	0.041	39
	(0.178)	(0.189)	(0.163)	

Each row re-estimates the riverine-versus-cropping interaction from the bottom panel of Table 5 at the row's radius in place of the main 30 km radius, with the same control set and fixed effects. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

B Data and Descriptive Statistics

B.1 Demographic Health Survey Sample

Table 36 reports summary statistics for the 7,603 children with valid anthropometry in the 2022 DHS. The anthropometric outcomes carry Kenya's general burden. Mean height-for-age is −0.91 with a standard deviation of 1.31, and a quarter of the sample sits below −1.77. Weight-for-height is closer to the reference median at −0.21, consistent with a population where chronic stunting is more prevalent than acute wasting. Children average 14.7 months at measurement and the sample is 49 percent female. Two thirds of children are rural, 48 percent own land, and 68 percent own livestock, but only 16 percent report an agricultural primary occupation, which is why the occupation variable enters only as a heterogeneity dimension and never as a control.

Distance to a major road is heavily right-skewed. The median child lives 2.2 km from one and the 75th percentile 5.8 km, but the maximum exceeds 114 km. Gestational SPEI averages 1.06 at six months, reflecting that the invasion window overlapped a period of above-average moisture in much of the country.

Table 36: Summary statistics, 2022 DHS children with valid anthropometry

	Obs.	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
Anthropometry								
Height-for-age (HAZ)	7,603	−0.91	1.31	−5.90	−1.77	−0.92	−0.11	5.95
Weight-for-age (WAZ)	7,600	−0.64	1.20	−5.29	−1.43	−0.63	0.13	4.85
Weight-for-height (WHZ)	7,595	−0.21	1.20	−5.00	−1.01	−0.20	0.58	4.82
Child and mother								
Age (months)	7,603	14.71	7.26	1.00	9.00	14.00	21.00	30.00
Female	7,603	0.49	0.50	0.00	0.00	0.00	1.00	1.00
First born	7,603	0.26	0.44	0.00	0.00	0.00	1.00	1.00
Twin	7,603	0.03	0.17	0.00	0.00	0.00	0.00	1.00
Birth order	7,603	3.23	2.24	1.00	1.00	3.00	4.00	14.00
Mother's education (yr)	7,603	7.66	5.08	0.00	3.50	8.00	12.00	24.00
Mother's height (cm)	7,603	160.76	4.39	139.70	160.40	160.70	161.00	180.40
Household and location								
Household size	7,603	5.90	2.57	2.00	4.00	5.00	7.00	24.00
Rural	7,603	0.67	0.47	0.00	0.00	1.00	1.00	1.00
Owns land	7,603	0.48	0.50	0.00	0.00	0.00	1.00	1.00
Owns livestock	7,603	0.68	0.47	0.00	0.00	1.00	1.00	1.00
Agricultural occupation	7,603	0.16	0.36	0.00	0.00	0.00	0.00	1.00
Distance to road (km)	7,603	7.28	15.65	0.00	0.74	2.18	5.82	114.65
Gestational window								
SPEI (6 month)	7,598	1.06	1.07	−2.11	0.25	1.20	1.98	3.06
NDVI z-score	7,603	0.69	0.80	−3.04	0.24	0.72	1.19	6.04
Maize price index	7,603	0.95	0.15	0.66	0.83	0.95	1.06	1.58

Mother's height is filled with the median among measured women for children whose mother was not measured (Section 4), which is why its interquartile range is narrow.

Figure 11 maps the 2022 cluster locations against the major-road network and the World Bank markets used for the price analysis. Clusters follow population density, dense through the western highlands, the central highlands, and the coastal strip, and sparse across the arid north and northeast. The road and market layers follow the same pattern, which is the source of the remoteness gradient that recurs throughout the results.

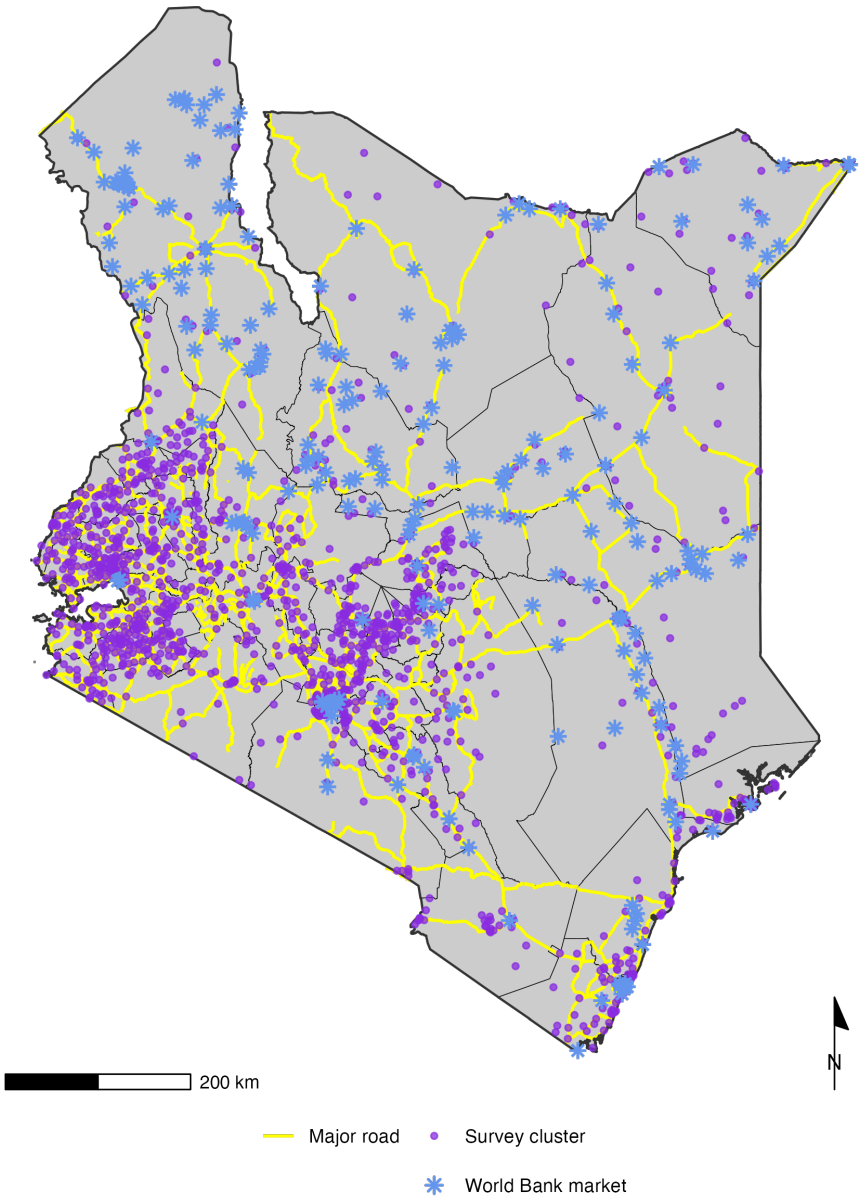


Figure 11: 2022 DHS cluster locations, major roads, and World Bank market locations.

Figure 12 marks the clusters with at least one child treated under the verified measure at 30 km. Treated clusters sit in the western and central highlands, the eastern slope and the coast, the cropping areas where the stage condition can hold, while the pastoral zones of the north and northeast hold none; the few treated clusters there lie in the riverine, agropastoral and cropping zones inside that region.

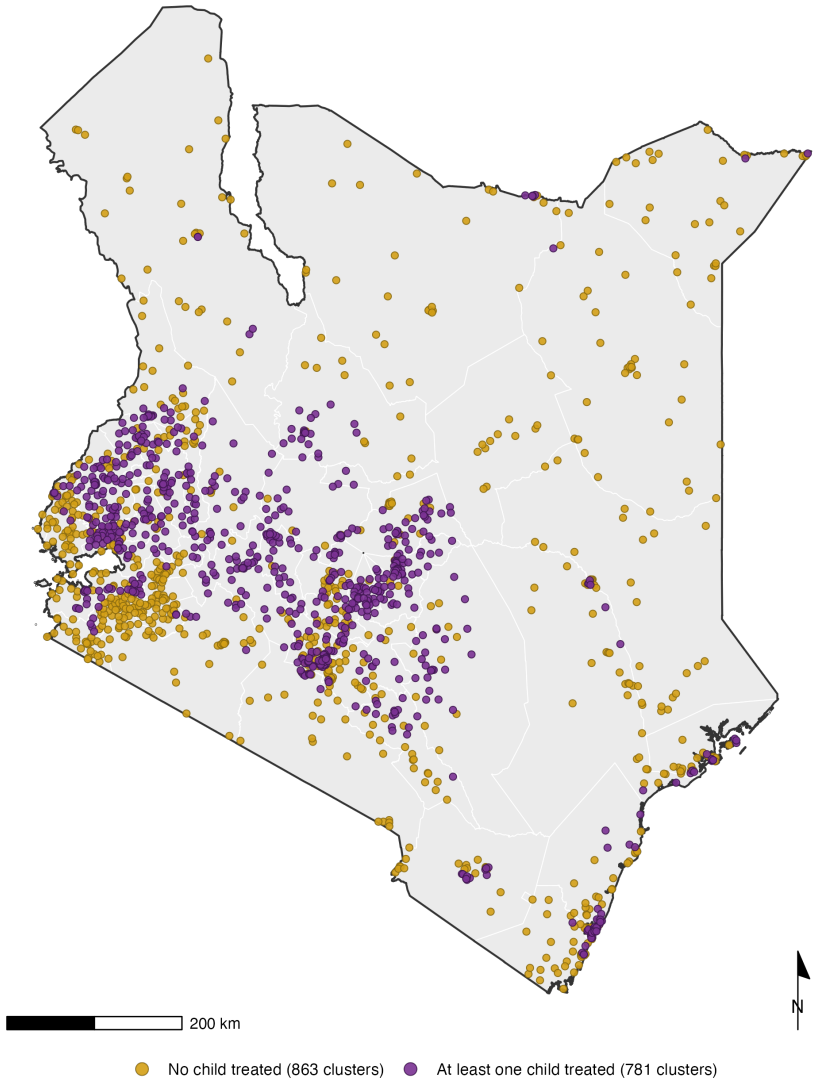


Figure 12: 2022 DHS clusters with at least one child treated under the verified measure at 30 km (purple) and with no treated child (mustard). Clusters are those with at least one child with valid anthropometry.

B.1.1 Sampling Weights

Figure 13 plots the distribution of DHS sampling weights on a log scale. The distribution is right-skewed, with the 99th percentile 6.3 times the median and the maximum 14.2 times, which motivates the winsorized robustness check in Table 37. Winsorizing weights at the first and 99th percentiles and re-running the full grid of 108 regressions leaves the median estimate at -0.074 against -0.090 untrimmed, the negative share at 92 percent against 96 percent, and the significant share at 24 percent against 27 percent. Sample size

and composition are identical under winsorizing, since weights are capped rather than observations dropped. Panel B of Table 37 reports the verified estimate at 10, 30 and 50 km with DHS weights, winsorized weights and no weights. Winsorizing shrinks the weight-for-height estimate at 30 km from -0.185 to -0.164 and dropping the weights shrinks it to -0.103, both still significant at the 5 percent level. The unweighted estimates are smaller on every outcome, so the effect is larger among the children the weights scale up. I report the weighted estimates as the main results because they describe the population of Kenyan children rather than the sample as drawn.

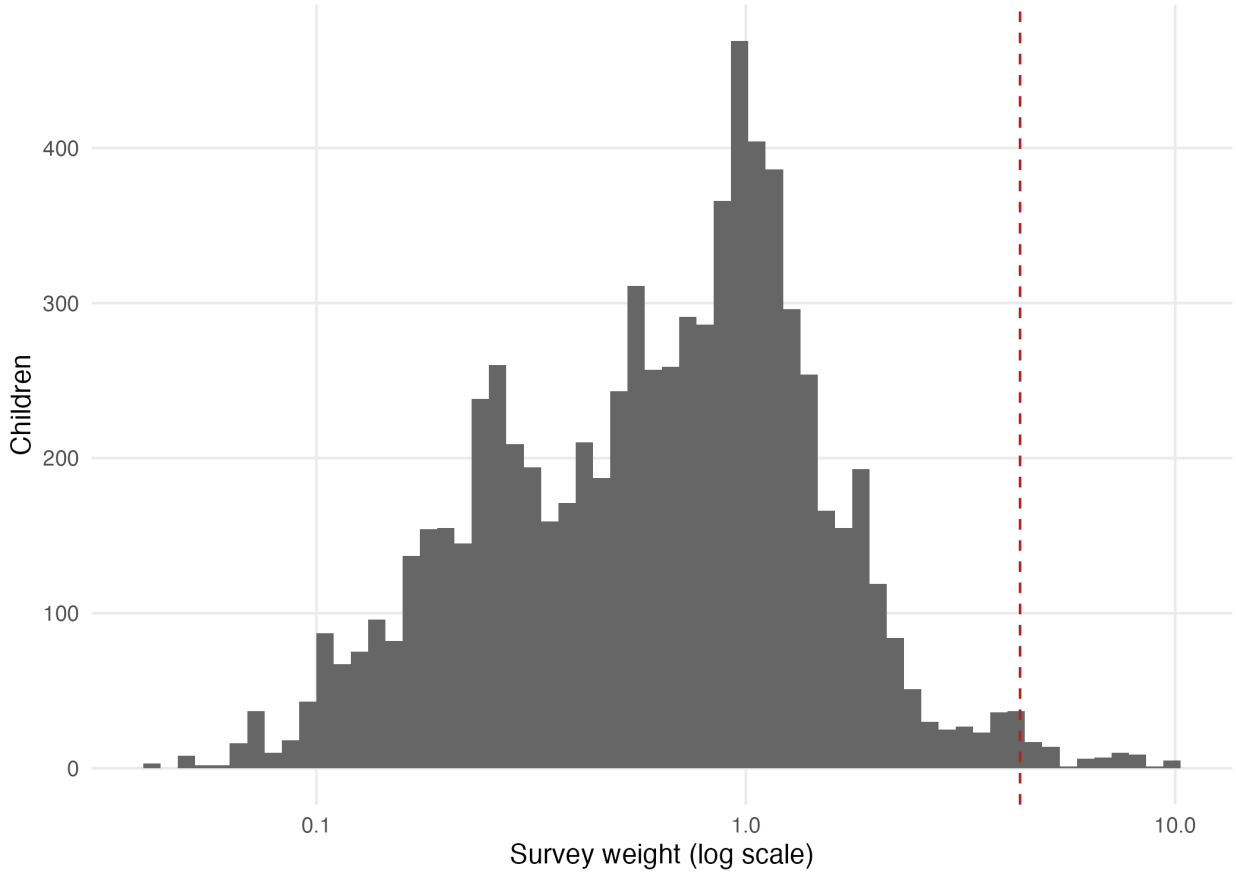


Figure 13: Distribution of DHS sampling weights. Dashed line marks the 99th percentile.

Table 37: Estimates under DHS, winsorized and no sampling weights

	DHS weights	Winsorized weights	Unweighted
Panel A: Core grid of 108 regressions			
Median estimate	−0.090	−0.074	—
Mean estimate	−0.095	−0.077	—
Share of estimates that are negative	0.96	0.92	—
Share significant at the 5 percent level	0.27	0.24	—
Panel B: Verified exposure			
Weight-for-height z-score			
10 km	−0.313***	−0.295***	−0.248***
	(0.084)	(0.076)	(0.065)
30 km	−0.185***	−0.164***	−0.103**
	(0.069)	(0.059)	(0.050)
50 km	−0.119*	−0.101*	−0.077
	(0.062)	(0.055)	(0.049)
Height-for-age z-score			
10 km	−0.193*	−0.165*	−0.068
	(0.099)	(0.085)	(0.062)
30 km	−0.059	−0.046	0.016
	(0.060)	(0.056)	(0.052)
50 km	−0.046	−0.032	0.021
	(0.056)	(0.051)	(0.050)
Weight-for-age z-score			
10 km	−0.337***	−0.309***	−0.222***
	(0.103)	(0.087)	(0.060)
30 km	−0.167**	−0.145**	−0.063
	(0.069)	(0.058)	(0.046)
50 km	−0.114*	−0.093*	−0.041
	(0.061)	(0.052)	(0.041)

Panel A summarizes the grid of four exposure definitions (proximity, damaging form, damaging form and crop stage, verified damage), nine radii from 10 to 50 km, and three anthropometric z-scores. Panel B reports the verified-exposure estimate at 10, 30 and 50 km. Each regression shares the control set and fixed effects of column (3) of Table 2. Winsorized weights cap the DHS individual sampling weight at its first and 99th percentiles; the unweighted column drops the weights. The sample is identical in every column. Standard errors clustered on a half-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

B.2 Climate and Vegetation Measures

Rainfall enters through CHIRPS at two temporal resolutions. The monthly product supports the gestational rainfall controls and the SPEI construction. The dekadal product, at ten-day resolution, supports growing-season onset detection. Onset cannot be identified from monthly totals, since a season that begins in the third dekad of March and one that begins in the first dekad of April differ by roughly ten days in the field but fall in adjacent calendar months, and the crop calendar shift described in Section 4.1.2 depends on resolving that difference. Extraction is a buffer mean over the DHS displacement radius rather than a point value.

SPEI is constructed at three- and six-month accumulation windows from CHIRPS precipitation and TerraClimate potential evapotranspiration, following Vicente-Serrano et al. (2010). The index standardizes the climatic water balance, precipitation minus PET, against that location's own historical distribution, so a value of 1.0 means the gestational window was one standard deviation wetter than normal for that cluster rather than wetter than some national average. Only the six-month index enters the control set. The three- and six-month indices are highly correlated within cluster and including both adds collinear variation without adding information, so the longer window is retained as the one more likely to capture a full growing season.

Vegetation enters through MODIS MOD13Q1, a 16-day composite at 250 m resolution. Two features of this product govern how it is used. The compositing window means the date of an NDVI observation is not the date of the underlying satellite pass, since the product selects the best available pixel within each 16-day period. This is why the pre- and post-arrival windows in equation (6) are defined in composites, and why the post-arrival window spans roughly 16 to 48 days depending on where within its composite an arrival falls. Pixels also carry a reliability flag. Pixels flagged as cloud- or snow-contaminated are dropped before averaging to the cluster, so a composite with heavy cloud cover contributes fewer pixels rather than contributing corrupted ones.

NDVI is converted to a per-cluster, per-composite z-score against that cluster's own pre-invasion climatology for the same composite of the year. This removes both the level difference between a humid highland cluster and an arid northern one and the seasonal cycle within each. A z-score of zero means vegetation was at its normal level for that place at that point in the year. The damage condition in equation (8) then tests a change in this z-score rather than its level, so a cluster that was already dry relative to its own norm does not qualify on dryness alone.

B.3 Crop Calendars

Two sources fix the crop calendar, as Section 3.4 describes. This appendix reproduces an example of each, along with the transcribed calendar those sources are turned into for the crop this paper discusses most, maize.

Figure 14 shows the FEWS NET seasonal calendar for Livelihood Zone 19, the Central Highlands High Potential Zone, which covers the high-rainfall counties north of Nairobi. The zone runs a single extended long-rains season. Land preparation for the maize and bean crop occupies March and April, planting runs from late April into June, and harvesting begins in October and continues to January. Irish potato and cabbage run two cycles. Coffee and tea are harvested across most of the year, and a footnote lists permanent crops including avocado, mango, macadamia, banana, and pineapple. Food prices peak in May and June, in the gap between planting and harvest, and market access is poor during the rains. A swarm arriving in this zone in May or June meets the maize crop at planting, rank 2 in equation (3). A swarm arriving between

October and January meets the crop under its harvest bar, rank 3. A swarm arriving in February meets bare ground, and one arriving between July and September, when the maize is growing but the calendar shows neither planting nor harvest, counts as out of season.

This is a limitation of the source. The FEWS NET calendars mark land preparation, planting and harvest, and leave the months between the end of planting and the start of harvest blank. A crop growing in those months stands in the field and can be eaten, but the calendars give no information on when a growing stage begins or ends for each crop and zone, and I do not interpolate one into the gap. Arrivals during the gap therefore count as out of season. They are excluded from the treatment and enter the out-of-season placebo group instead, so to the extent that they destroyed growing crops, treated children sit in the comparison group and the estimates are biased toward zero.

Figure 15 shows the calendar for Livelihood Zone 9, the Northeastern Pastoral Zone, covering Wajir and the surrounding rangeland. The calendar has no crop section. It records camel calving, kidding and lambing, disease outbreaks, milk availability, migration, livestock sales, and the lean season. Households here obtain food from milk, livestock sales, and purchase, and the FEWS assessment calendar tracks livestock condition rather than crop condition. No date in this zone places a field in a susceptible stage, so equation (5) never holds and no child in this zone is treated. Section 5.4 reports what proximity exposure predicts in these zones, which is nothing.

Figure 16 collects the maize calendar transcribed from calendars like Figure 14, one row per livelihood zone. Zones without a maize row, such as the pastoral zone in Figure 15, are absent from the figure rather than shown blank. The land preparation and planting window moves with each zone's rains, so the bimodal zones on the coast and around Lake Victoria show two separate cropping windows in a single calendar year while the single-season zones, including the Central Highlands zone above, show one.

Seasonal Calendar: Livelihood Zone 19 - Central Highlands, High Potential Zone

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Seasons	**Dry**		**Long rains**									**Dry**
Crops												
Legend	Land Preparation				Planting				Harvesting			
Maize + Beans												
Coffee												
Tea												
Irish potato												
Cabbage												
Pyrethrum	Include pruning											
Others*	**Grown and harvested at different times throughout the year.**											
Milk availability	**Throughout the year due to dairy farming**											
Livestock	**Livestock Husbandry, vaccinations and general management throughout the year (intensive)**											
Food prices					High							
Market access			Poor							Poor		
Labor		Peak										
Crop Assessments												

* Permanent crops including avocado, passion fruit, mangos, macadamia nuts, bananas and pineapples

Figure 14: FEWS NET seasonal calendar, Livelihood Zone 19, Central Highlands High Potential Zone.

Seasonal Calendar: Livelihood Zone 9 - Northeastern Pastoral Zone

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Seasons	**Dry**			**Long rains**		**Dry**				**Short rains**		
Livestock production												
Camel Calving											Peak	
Kidding/lambing				Peak							Peak	
Disease outbreaks						Highly likely						
Prices				Peak							Peak	
Milk availability												
Camel milk												
Migration		Dry		Wet			Dry				Wet	
Others												
Livestock sales	High					High			High			High
Risk of insecurity				High			High					High
Malnutrition								High				
Lean season												
Labor Availability		Peak							Peak			
Market access				Poor						Poor		
Water stress												
Cross border inflows						Peak						
Food price					High			High				
FS Assessments												

Figure 15: FEWS NET seasonal calendar, Livelihood Zone 9, Northeastern Pastoral Zone. The calendar contains no crop section.

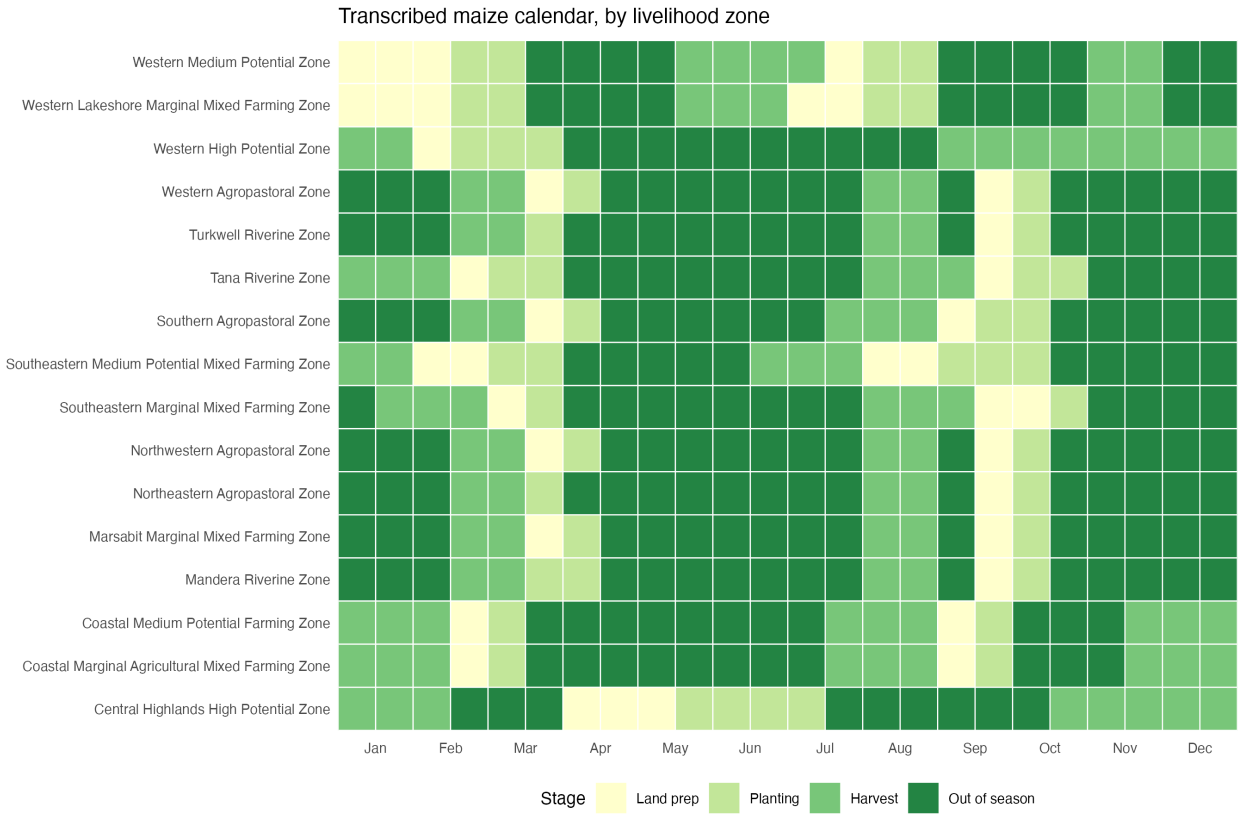


Figure 16: Transcribed maize calendar, by livelihood zone, built from calendars such as Figure 14.

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